

**Rochester Institute of Technology
Department of Computer Engineering
CMPE-630 Digital IC Design
Spring 2025
Laboratory Exercise 8
Auto Place and Route Tutorial**

Due: 04/04/2025 end of lab

Contents

1	Introduction	3
2	Block Place and Route Design Flow	3
3	Create a Hierarchical Block Schematic	5
4	Create Initial Hierarchical Block Layout	7
5	Configure Hierarchy	8
6	Placement Strategy	14
7	Generate Layout From Source	20
8	Changing Pins	24
9	Placement Planning	25
10	Place Design	29

11 Optimize Pin Placement	33
12 Power Strapping	38
13 Wire Planning	42
14 Routing	48
15 DRC	50
16 LVS	51
17 Run Parasitic Resistor, Capacitor(RC) Extraction (RCX)	51
18 Part 2: Analog — Digital Mixed-Signal Simulation	52
19 Sign-off Sheet	70

1 Introduction

This tutorial consists of two parts, the first part will introduce you to the version 6.x Virtuoso Layout Suite GXL design environment from Cadence Design Systems. You will target a generic 45nm process design kit (PDK) and work through the various steps in the hierarchical design flow to create a block level design: schematic capture, layout generation from source, placement planning, automatic placement, pin optimization, automatic routing, design rule checks, and parasitic extraction.

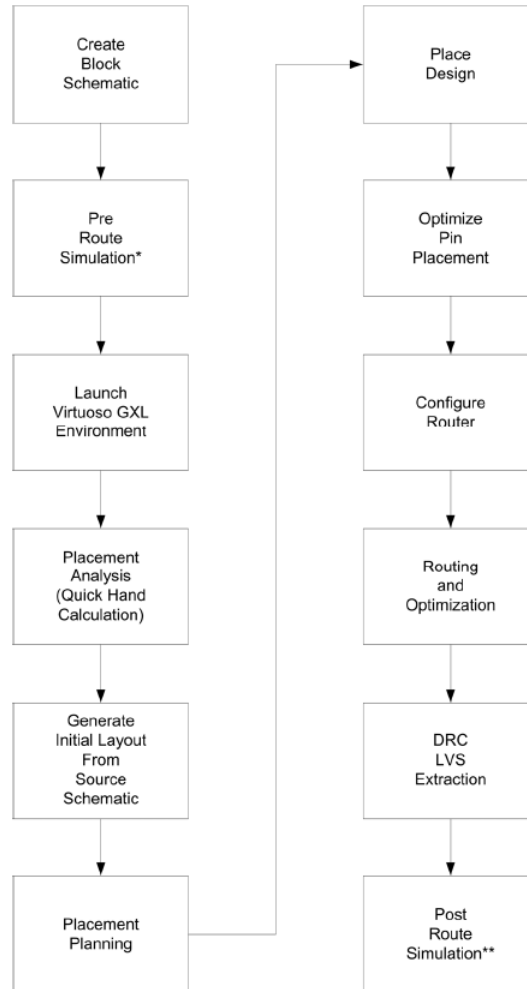
For the second part of this Tutorial you will be introduced to the version 6.x AMS Designer Simulation environment from Cadence Design System. You will perform mixed signal simulation of the block level design. Strict adherence to design methodology **REQUIRES** you to simulate a block to verify that your circuit works **pre** and **post** layout.

As you work through this Tutorial you should enjoy the level of integration and automation in this design flow.

A Friendly Suggestion Before Getting Started: Many of the features of the software can be accessed via icons (for the most common tools), or menu picks; many of the menu picks also include a keyboard key shown as part of the menu palette. Pressing the appropriate keyboard key is a shortcut (called a bind key) to that function. It is strongly suggested that you **learn bind keys**. The software is also context sensitive and remembers your last command (perhaps as a productivity feature). This can often be annoying; the ESC key will exit the current context.

2 Block Place and Route Design Flow

Figure 1 introduces the block place and route design flow for the final project. As noted earlier, the Tutorial is broken in two parts (i) Part 1 for the Place and Route (P&R); (ii) Part 2 for the pre-route mixed signal simulation. Post-route simulation will not be performed during this Tutorial. However, that post-route simulation **MUST** be performed during a proper block design process.



* - Task performed during Part 2 of EDA Tutorial 2

** - Task not performed during EDA Tutorial 2

Figure 1: Block Place and Route Design Flow

3 Create a Hierarchical Block Schematic

Logon, return to your cmpe530/630 directory and restart the Cadence tools.

In the Library Manager window, select the library you created in Lab Exercise 4, and then select File → New → Cellview. Give your cell schematic the name **FML_ADD8** (where FML stands for your First, Middle and Last initials).

Now, using the Mirror Adder from the tutorial done in Lab Exercise 6, cell **FML_MIRRORADDER**, design the schematic for the circuit shown in Figure 2 below.

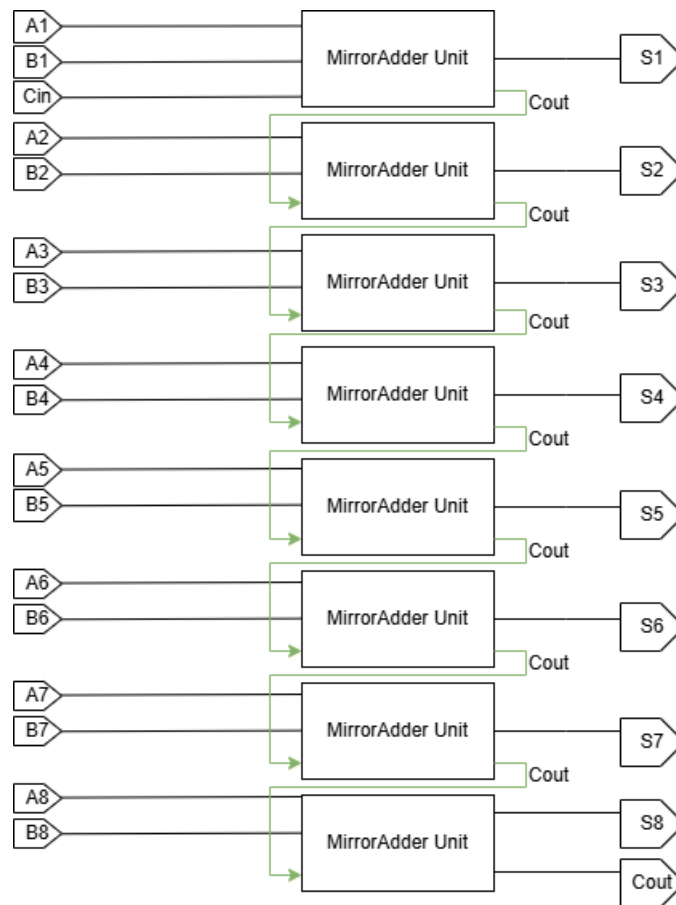


Figure 2: Reference Schematic

Once complete, your schematic should look similar to Figure 3

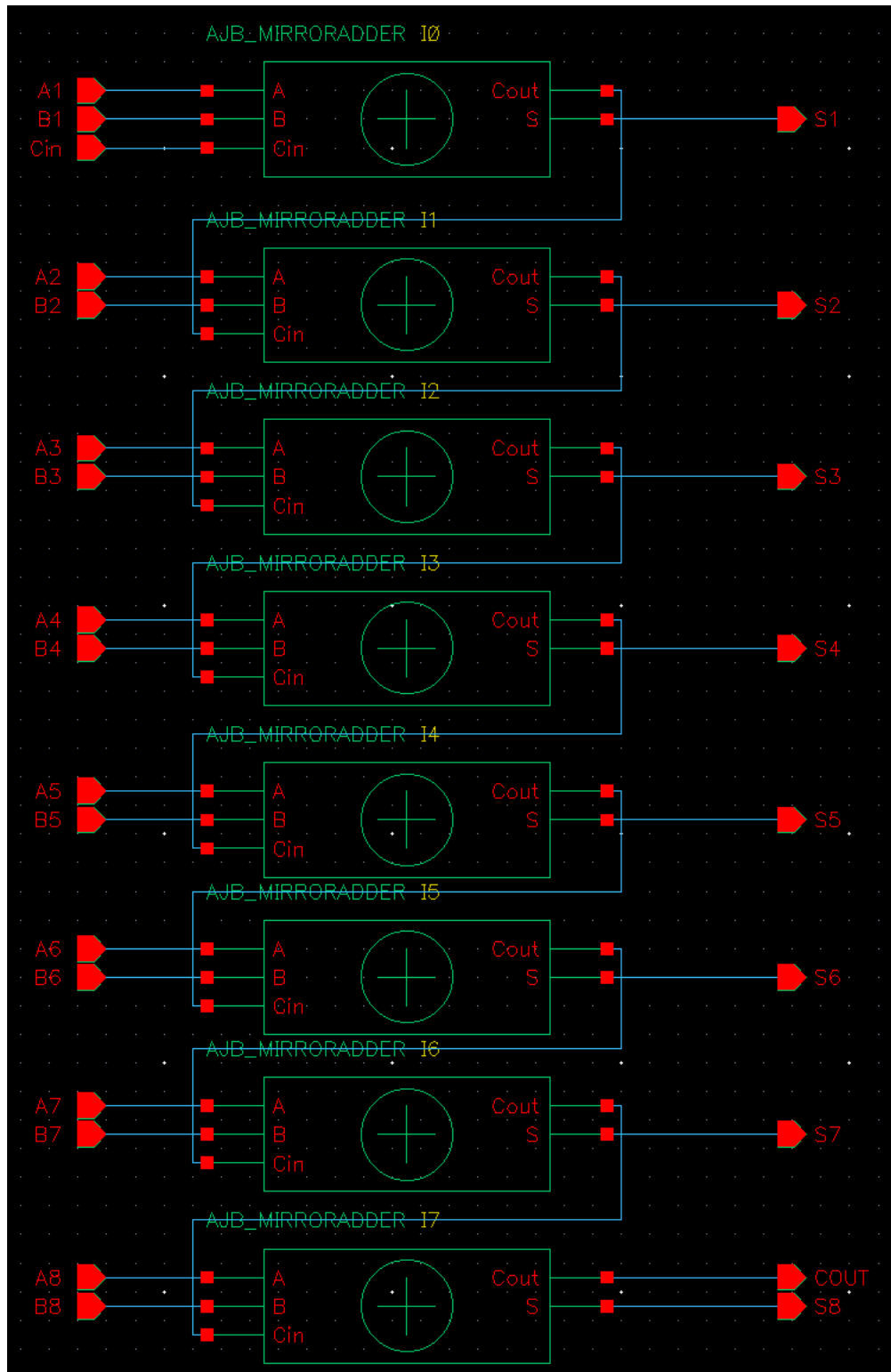


Figure 3: FML_ADD8 Schematic

4 Create Initial Hierarchical Block Layout

From the schematic window, select Launch → Layout GXL as shown in the Figure 4



Figure 4: Launch Virtuoso Layout GXL

The following physical implementation Startup Options window appears; configure as shown.

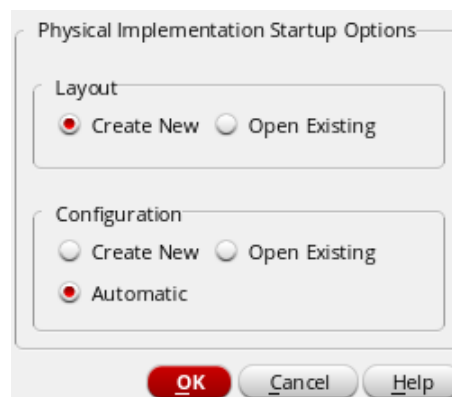


Figure 5: Physical Implementation Startup Options Window

Click OK. The following new file window will appear configure as shown. Note that the View is

layout and the Type is layout.

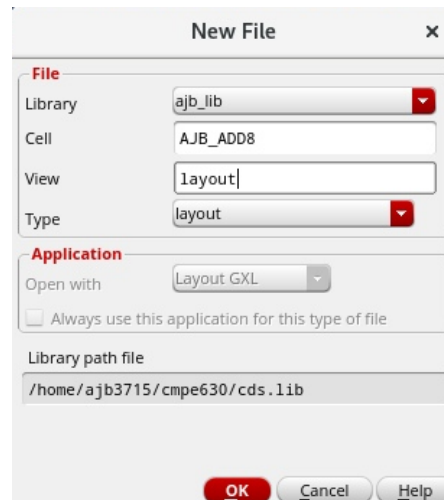


Figure 6: Physical Implementation New Layout File Options Window

Note that later screenshots of the lab may have the name "layout_new" when referring to the layout, ignore this naming convention and just simply name the file "layout". Click OK.

Due to the tight coupling between the schematic and layout in Virtuoso Layout GXL, the tools will continue your windows. After this, a popup may appear regarding the Physical Hierarchy, press OK and you will load into a window similar to Figure 10, close this window for now.

5 Configure Hierarchy

Due to the nature of our library, we need to perform a quick configuration to map out cells to cell sites (physical geometry information the place and route tools understand). Cell sites are defined in a README that's included with the Generic 45nm standard cell technology library. **Using the Library Manager access the README as shown below.**

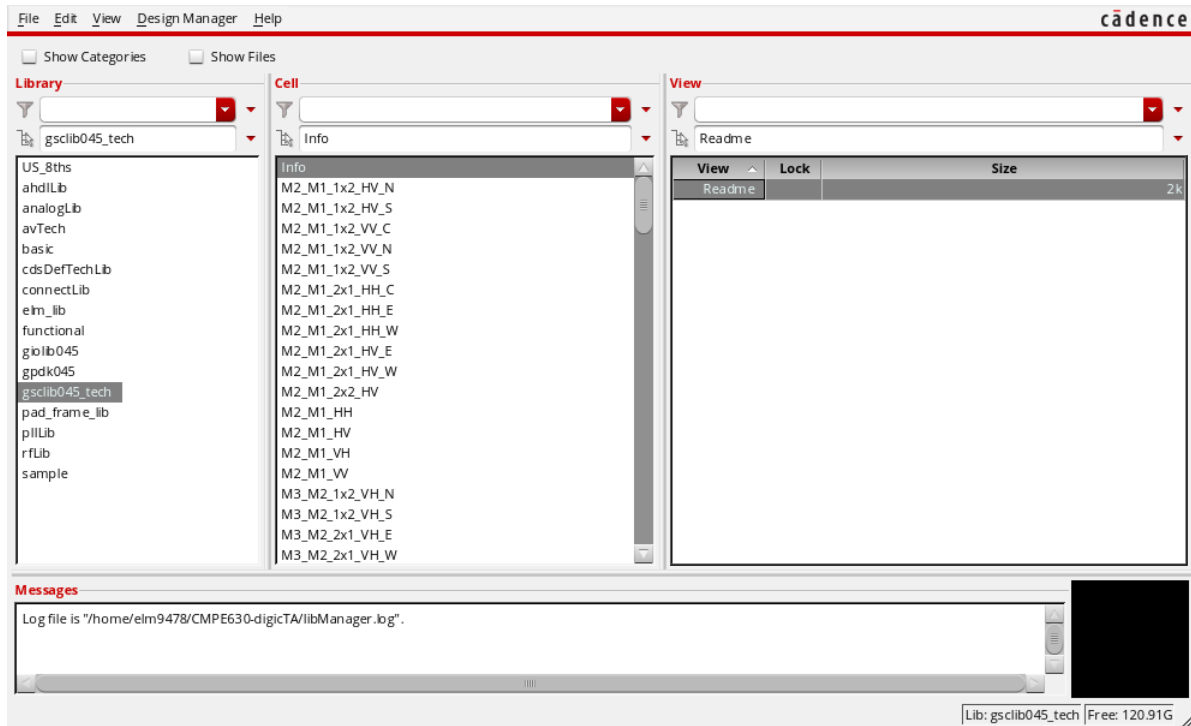


Figure 7: Library Manager Window

Open the README file, then you will see a text editor window appear as shown in Figure 8.

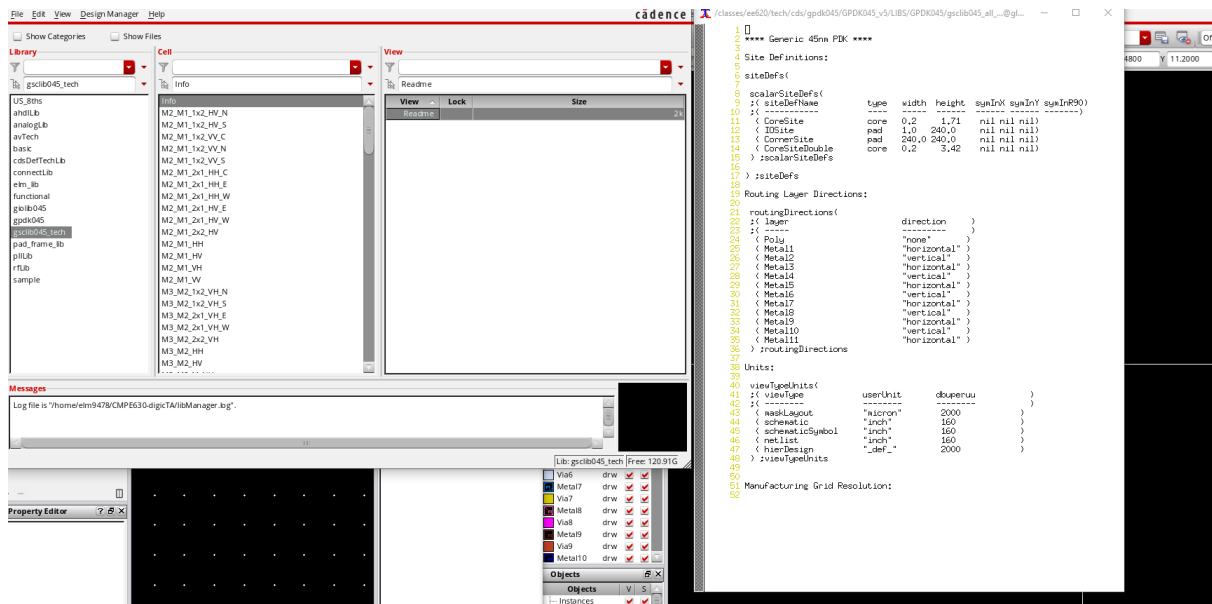


Figure 8: Generic 45nm Standard Cell Technology Library README File

Note that this file contains information regarding cell sites, routing, grid, etc.; you will need to reference this file more than once throughout the place and route process. Review the README to inspect the information provided.

Also note that the contents of the README File may vary person to person.

Now select the Launch → Configure Physical Hierarchy as shown in Figure 9 below.

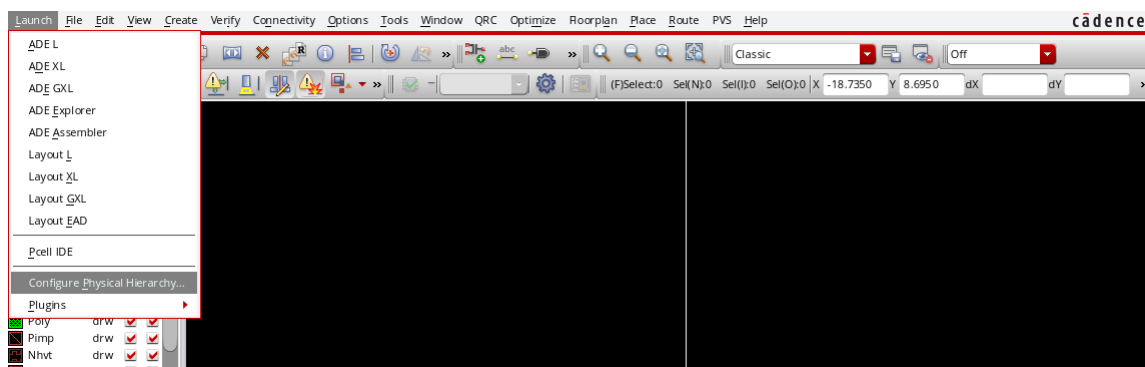


Figure 9: Launch Configure Physical Hierarchy

The initial Configure Hierarchy window appears. Now select the **Component Types** as shown below.

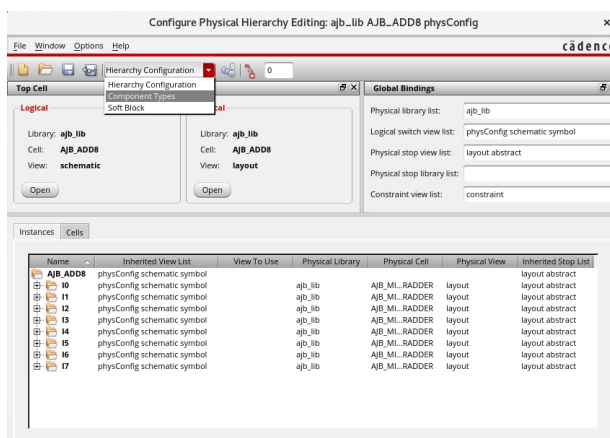


Figure 10: Configure Hierarchy Window

Note that your Mirror Adder cell does not currently have component type binding.

Right click on your Inverter cell and select the menu option **Add component type** as shown below.

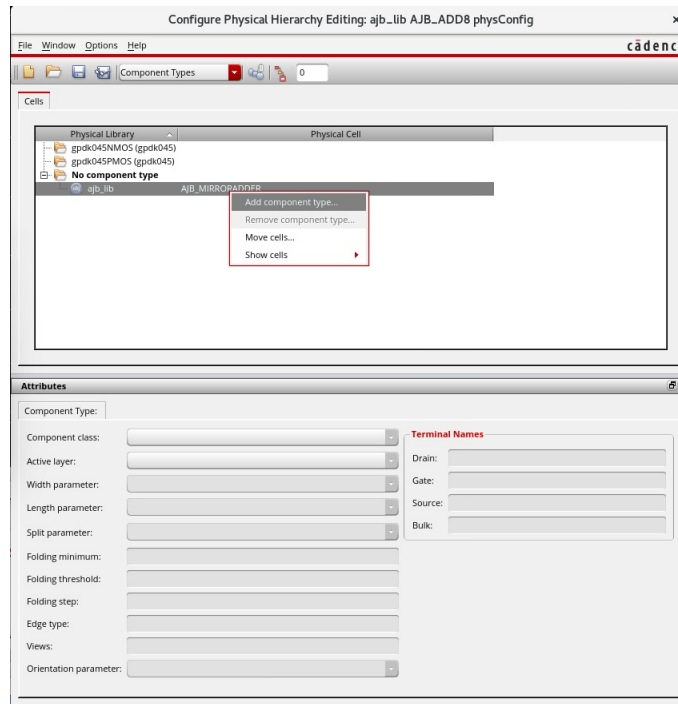


Figure 11: Configure Hierarchy Window

A new window appears as shown below.

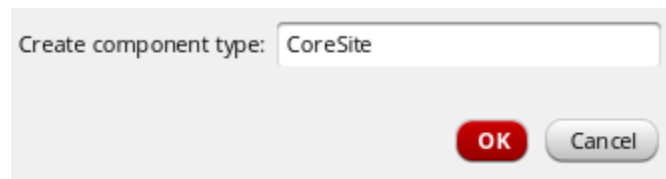


Figure 12: Create Component Type Window

Enter the component type CoreSite (as shown in the library README under siteDefs → scalarSiteDefs) and click OK.

We need to identify the class of components bound to our new CoreSite component type. Select CoreSite component type.

In the **Attributes** section, select the **STDCELL** Component Class.

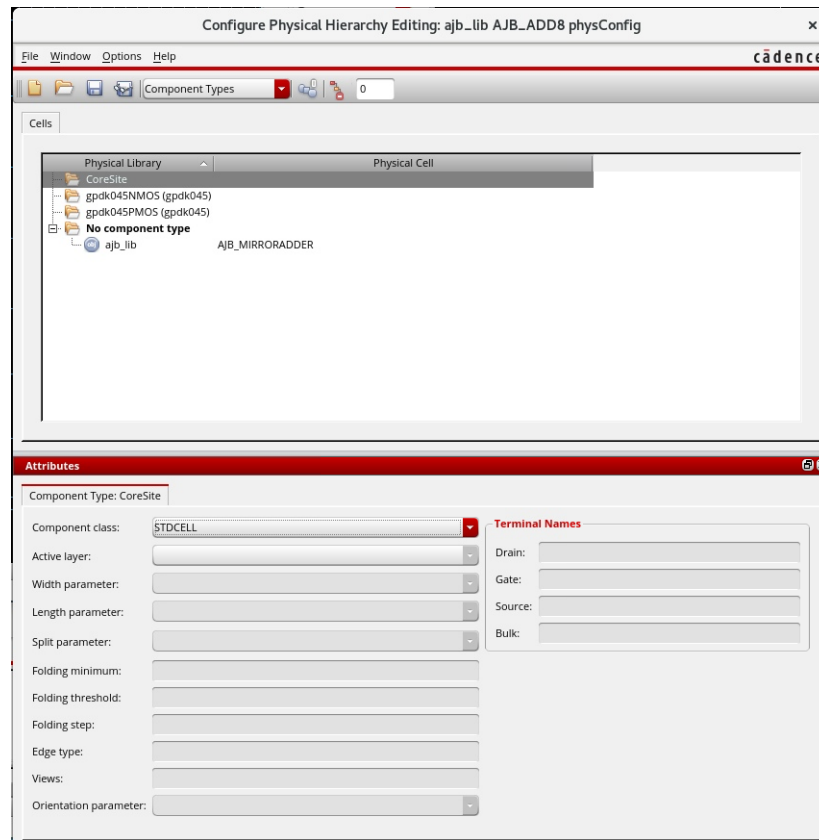


Figure 13: Configure Hierarchy Window

With the CoreSite component type configured, let's move our cells to this component type. Right click on your Inverter cell and select the menu option **Move Cells** as shown below.

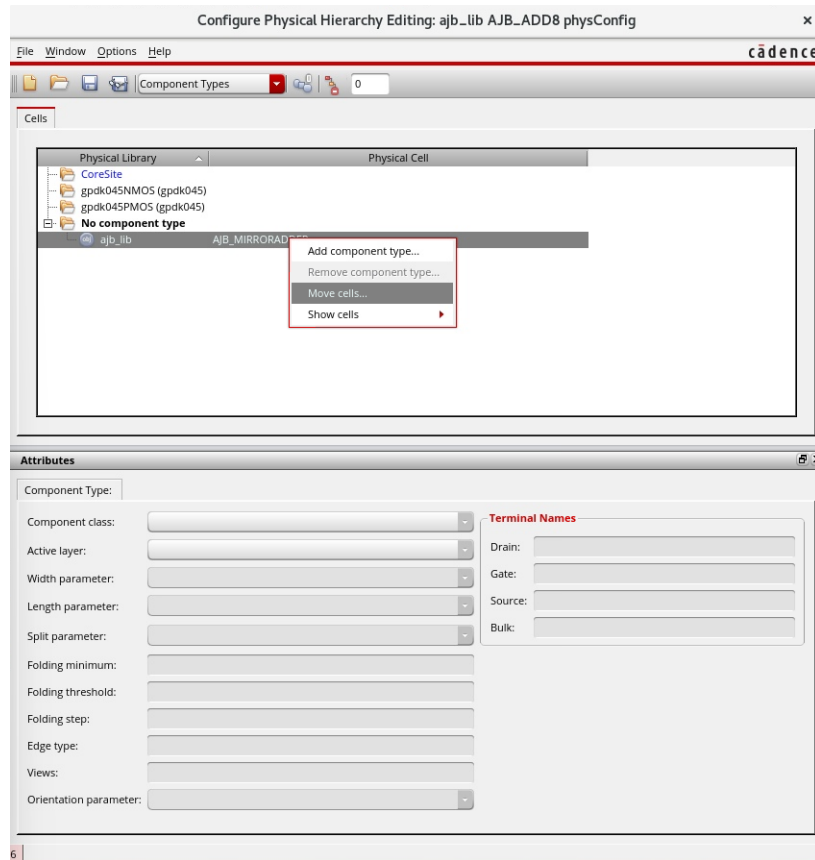


Figure 14: Configure Hierarchy Window

The following window will appear; make sure the CoreSite is selected.

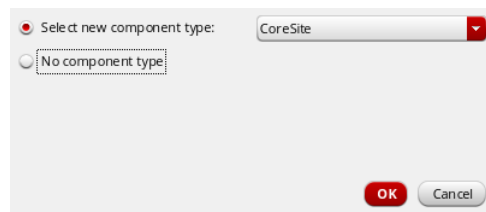


Figure 15: Configure Component Type Window

Click OK and your final window should appear as below.

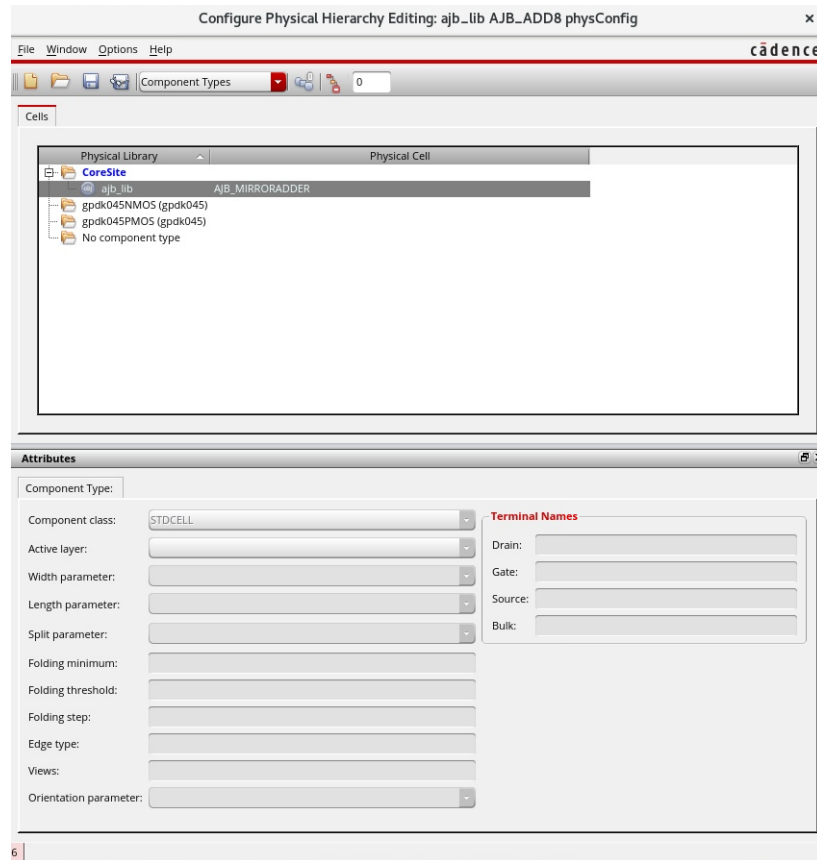


Figure 16: Configure Hierarchy Type Window

Click the Floppy Disk icon to save the configuration. Once saved you can close configuration form.

6 Placement Strategy

As noted earlier, effective use of place and route tools requires careful, upfront planning. The goal is to create a finished block level layout (see Figure 17 below as an example) that you can use within your chip level floor plan.

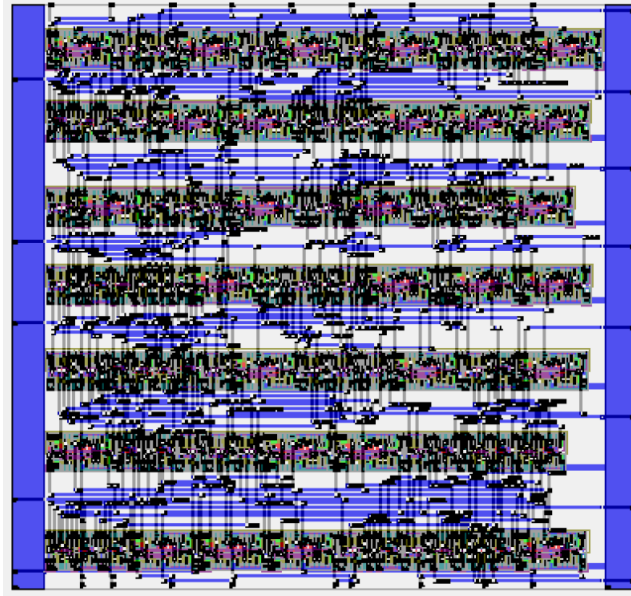


Figure 17: Finished Block Level Layout, 7 Standard Cell Rows, 8 Routing Channels, with Power Straps.

As you have carefully design your standard calls during one, developing a placement strategy should be a simple task.

Now perform some quick calculations on our design which is comprised of 8 copies of your FML_MIRRORADDER cell. We know the dimensions of this cell (we can measure dimensions from the layout), and that we chose λ to to be $25nm$ ($0.025\mu m$) for the Generic 45nm process.

Standard Cell Routing Grid Layout
 Note: All Dimensions are multiples of Lambda

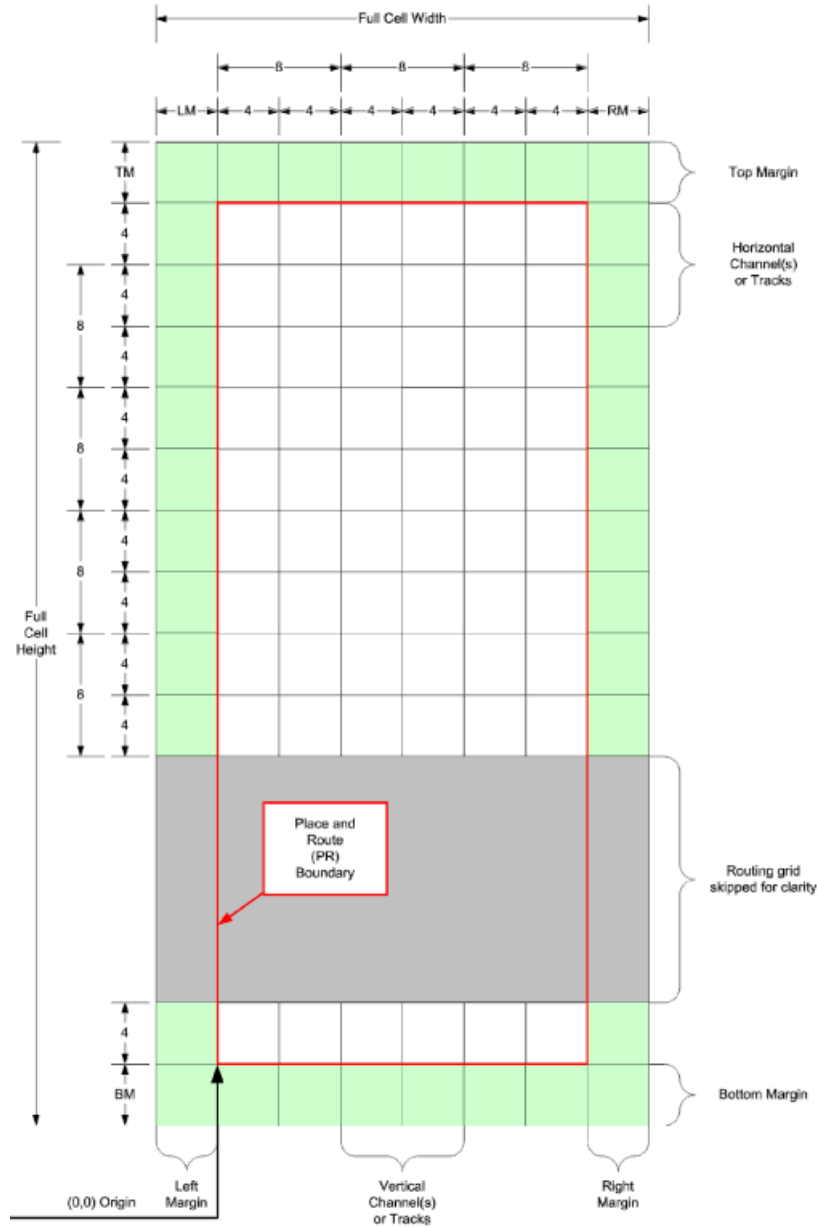


Figure 18: Example Cell with Routing Grid Overlay

Using the Virtuoso ruler tool, we obtain the following Cell Dimensions (recall that FML_MIRRORADDER) cell is 3 tracks wide, the example grid in Figure 20 is 3 tracks wide).

As shown in Figure 19 below, the idea is to create rows of cells instances (in this case, 2 rows will be sufficient) and surround the rows with a bounding box; we'll allow two tracks (recall that each track is 8λ) top and bottom, and left and right to create an area known as a routing channel between the PR Region and bounding box.

As defined, a row of standard cells will be placed within the PR Region, and pins will be placed at

the periphery of the Bounding box.

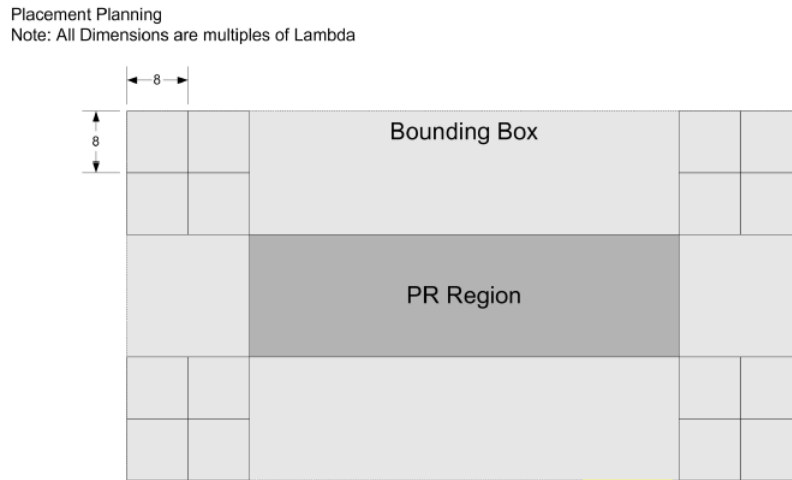


Figure 19: Placement Planning - PR Region Placed within the Bounding Box (not to scale)

As shown in Figure 20, we then calculate the items discussed above in terms of the total cell area, size of a standard cell row, the PR region that will contain our row cells, and finally the estimated size of the bounding box which will contain our now, IO and power pins.

If the clock requires more than one row, we'll define a space between rows (as multiples of tracks) and calculate the PR Boundary accordingly to account for the space between rows necessary as shown in Figure 21. In later sections in the lab, the upper value is utilized (0.8) despite only having 2 tracks. This is done to air on the side of caution, and possibly avoid unwanted errors.

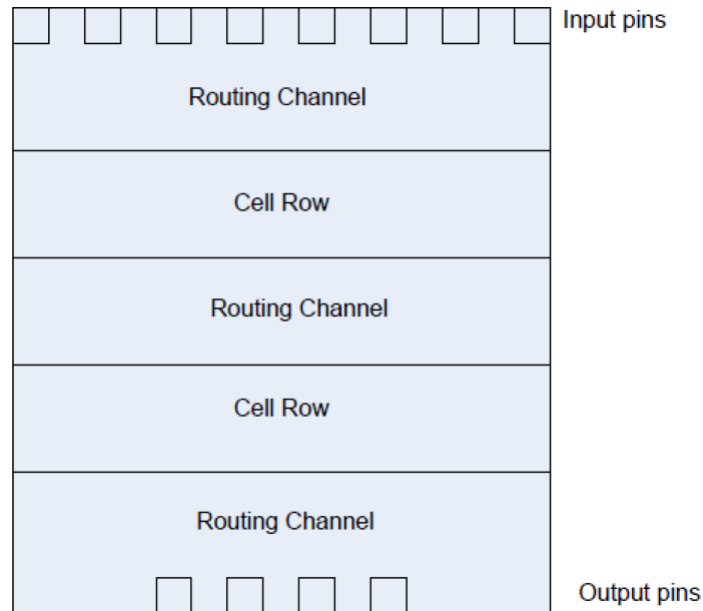
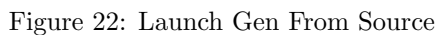


Figure 21: Example Cells Rows and Routing Channels

As shown in Figure 17, we'll also strap the power rails VSS on the right of the block, and VDD on the left of the block (or vice versa depending on the overall power grid of your chip). Note that left / right side power strapping should be **at least** 30λ wide.

Now that we have a placement strategy, let's generate an initial layout from the source schematic. Select Connectivity → Generate → all From Source as shown below.



Gen From Source automatically creates layout objects for the various components of your schematic, it also adds a Place and Route Boundary (PR Boundary). The PR Boundary is important as it sets a drawn area boundary of the layout cell. The various integrated place and route tools use the PR Boundary as an outer limit (similar to a perimeter fence): (i) layout object are placed within the PR Boundary; (ii) routing of wiring cannot past the PR Boundary.

The Gen From Source form has multiple tabs, start by configuring the Generate tab as shown in Figure 23. Note that we are including all items during generation (Instances, I/O Pins, and PR Boundary), but we are NOT placing them with the generated PR Boundary ('In Boundary' is not Checked). Now, configure the Generate parameters as shown in Figure 23

Generate

☒ Instances
☐ Chain ☐ Fold ☐ Chain Folds

☒ I/O Pins
☐ Except Global Pins
☐ Except Pad Pins

☒ PR Boundary
☐ Snap Boundary

Position

Minimum Separation: 0 ☐ In Boundary

Device Correspondence

☐ Preserve User-Defined Bindings

Connectivity Extraction

☒ Extract Connectivity after Generation

Figure 23: Gen From Source Form - Generate Options

Now configure the I/O Pins. Click on the I/O Pins tab. Here you will configure Gen From Source to create pins for the block level I/O. For the purposes of this Tutorial, we'll be placing the I/O pins on the top and bottom of the block. Being in the vertical direction, we'll define these pins using the Metal 4 — Pin layer. VDD and VSS will typically be strapped on the left and right sides (wiring out to the power strap in the horizontal direction), and we'll define these pins using the Metal 1 — Pin layer. The following figures show them being defined on the metal 3 layer, which can work as well but requires via placement to accommodate the power strapping that will occur layer. It is recommended to use metal 1.

Default Values for Electrical Pins

Layer: Metal4 pin Width: 0.08 Height: 0.08 Num: 1 Create: ☒ Apply

Specify Pins to be Generated

Select: Number Of Matches: 0 Add New Pin...

Term Name	Net Name	Layer	Width	Height	Num	Create
S4	S4	("Metal4" "pin")	0.08	0.08	1	t
S5	S5	("Metal4" "pin")	0.08	0.08	1	t
S6	S6	("Metal4" "pin")	0.08	0.08	1	t
S7	S7	("Metal4" "pin")	0.08	0.08	1	t
S8	S8	("Metal4" "pin")	0.08	0.08	1	t
VDD!	VDD!	("Metal3" "pin")	0.08	0.08	1	t
VSS!	VSS!	("Metal3" "pin")	0.08	0.08	1	t

Name: AS IS Layer: Metal3 pin Width: 0.08 Height: 0.08 Num: 1 Create: ☒ Update

Pin Label

☒ Create Label As ☒ Label ☐ Text Display Options...

Figure 24: Gen From Source Form - I/O Pins Options

Pin labels need to be created too, check **Create Label As: Label**, and click **Options**. The label form appears as shown below.

Figure 25: Gen From Source Form - I/O Pins Label Options

Once finished, click OK and you will return to the Gen From Source I/O Pin form. With the labels settings configured, you will adjust the target pin layer by selecting the pins in the listing window - VDD! and VSS! are selected and highlighted as shown in Figure 26.

Term Name	Net Name	Layer	Width	Height	Num	Create
S4	S4	("Metal4" "pin")	0.08	0.08	1	t
S5	S5	("Metal4" "pin")	0.08	0.08	1	t
S6	S6	("Metal4" "pin")	0.08	0.08	1	t
S7	S7	("Metal4" "pin")	0.08	0.08	1	t
S8	S8	("Metal4" "pin")	0.08	0.08	1	t
VDD!	VDD!	("Metal3" "pin")	0.08	0.08	1	t
VSS!	VSS!	("Metal3" "pin")	0.08	0.08	1	t

Figure 26: Gen From Source Form - I/O Pins Options

With the pins selected you want to modify, select the new Layer, confirm with the **Label Options**, then click **Update**. The selected pins should update to the chosen layer.

Now click on the **PR Boundary** tab. We'll enter the coordinates of the Bounding Box starting at the 0,0 origin, as shown below.

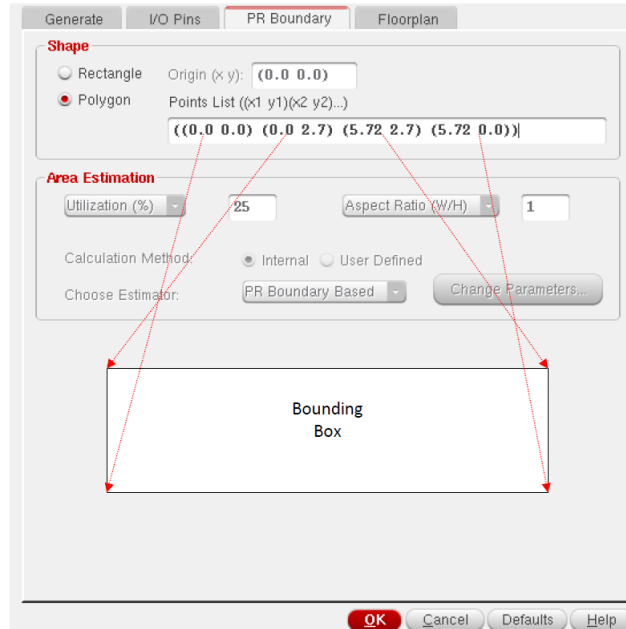


Figure 27: Gen From Source Form - PR Boundary Options. See Note below

Note that based on your calculations the Bounding Box coordinates may be different (although as instructed, start at the 0,0 origin). Although we define the PR Boundary as a rectangle, we could enter many more vertex locations and define a very complex polygon.

We don't have any hard macros or other floorplanning objects, so we can skip the Floorplan tab.

Click OK and Gen From Source will create the initial layout objects as shown below. **Note that your layout will look different from what is shown in the following figures.**



Figure 28: Gen From Source Generated Layout with PR Boundary

Note PR Boundary located at the 0,0 origin and check that your pins are on the intended layer. **If not, see the next section.**

8 Changing Pins

If your pins and labels are on the correct layers, you can skip to the next step; if not, read on.

Ideally, if you notice something wrong with your pins after Gen From Source you should START OVER - since you've done no meaningful work on the layout, for consistency of your layout database starting over is the recommended approach.

However, if you wish to make a quick adjustment such as changing VDD and VSS to be on Metal 3 — Pin, you will need to adjust both the PIN and LABEL layer. In this case you would select the appropriate VDD and VSS pins and labels and change them to be on layer Metal 1 — pin. Here are the steps.

1. Select the pin(s) you wish to edit (you can left click and drag to create a selection set).
2. Edit → Properties from the menu or “q” as a shortcut key.
3. Select the “Common” radio button on the right of the form so you can change all at once (just make sure you select all the right pins and labels), change the layer, and click OK. This will change the group.

9 Placement Planning

With our initial layout objects created, it's time to configure placement parameters. To do this we'll launch the digital placement planner, select Place → Custom Digital → Placement Planning as shown below (note that the image provides a preview of the final places standard cell row).

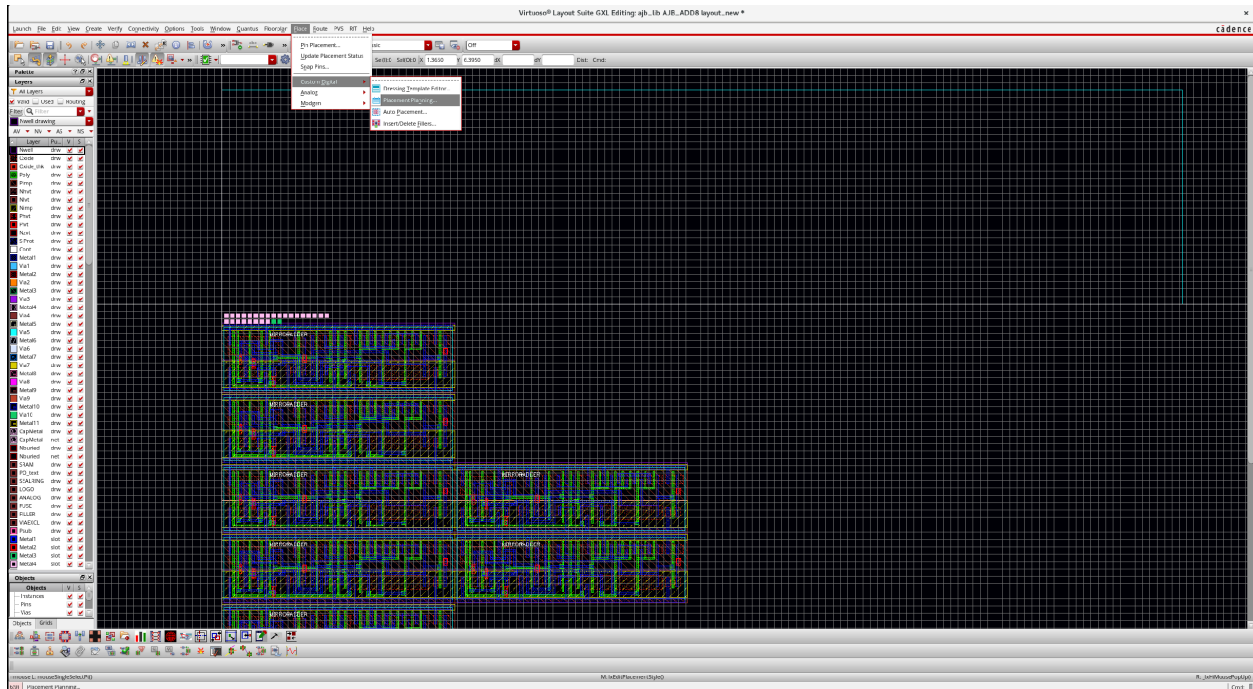


Figure 29: Launch Custom Digital Placement Planning

When the initial placement planning form appears, select **Create** and **Placement Region** as a **Rectangle**, and click **Draw**.

Placement Planning: ajb_lib AJB_ADD8 layout_new

Tap Cell Row **Create**

Rows ☒
Tap Cells ☐

▼ **Region**

☐ Boundary ☒ Placement Region

☒ Rectangle ☐ Rectilinear

Origin X Origin Y

Width Height

Draw Update Delete

► **Options**

Insert Delete Close Help

Figure 30: Custom Digital Placement Planning - Regions Options

Draw a rectangle within the PR boundary created during the previous Gen From Source step. Don't worry about the dimensions; we'll fix them using the Placement Planing **Regions** form. Just make sure the box you draw is within the PR Boundary.

Now modify the values of the Origin X, Origin Y, Width and Height based on the PR Region calculated earlier as shown in Figure 31. Click **Update** to update the PR Region.

Placement Planning: ajb_lib AJB_ADD8 layout_new

Tap Cell Row **Create**

Rows ☒
Tap Cells ☐

▼ **Region**

☐ Boundary ☒ Placement Region

☒ Rectangle ☐ Rectilinear

Origin X 0.4 Origin Y 0.33

Width 25.28 Height 5.02

Draw Update Delete

► **Options**

Insert Delete Close Help

Figure 31: Custom Digital Placement Planning - Regions Options

The PR Region will update as shown in Figure 32 below.

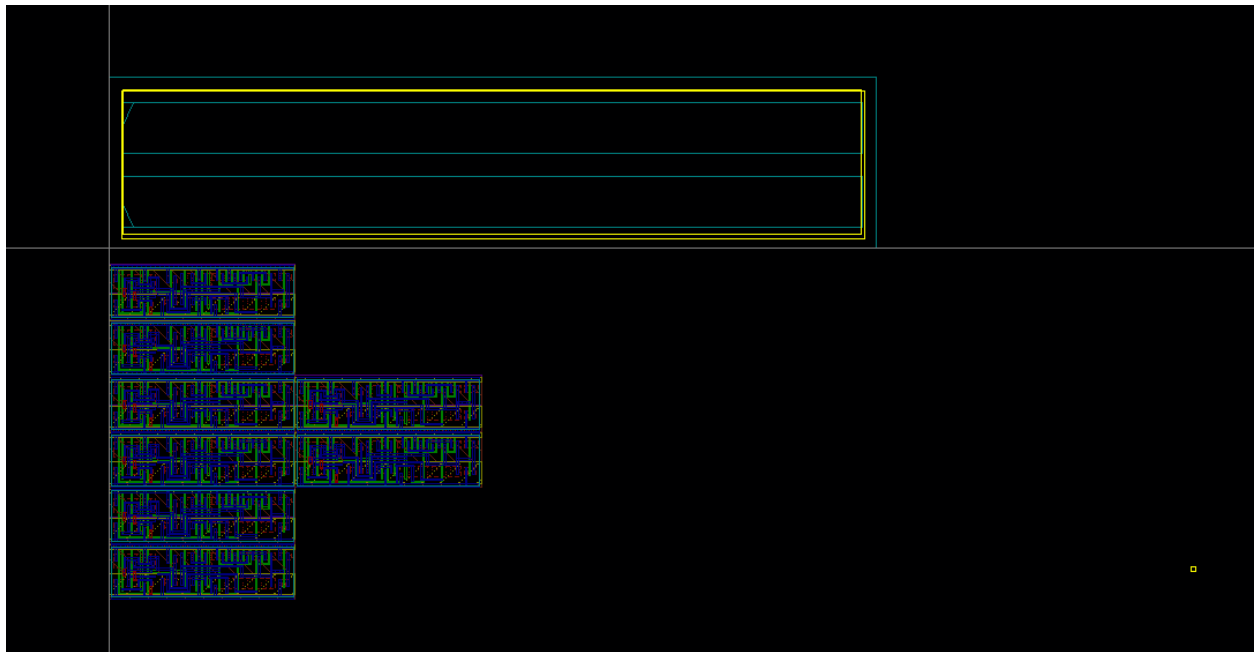


Figure 32: Placement Planning - Final PR Region

Select the **Rows** tab, and update the form as shown below including the power rails parameters.

Figure 33: Custom Digital Placement Planning - Rows Option

Note: (i) Make sure to specify Row Spacing as 0.8! If not done, the rows will overlap.

Note: (i) if VDD! does not appear in the Power Net dropdown, you must type VDD! into the Power Net field; (ii) if VSS! does not appearing the Group Net dropdown, you must type VSS! in to the Ground Net field.

Now click **Generate Rows**. Once complete, as shown in Figure 34 below, two rows should appear where the PR Region was defined with power rails. Check the Placement Statistics report in the CIW as shown in Figure 35. You can now close the Placement Planning form.

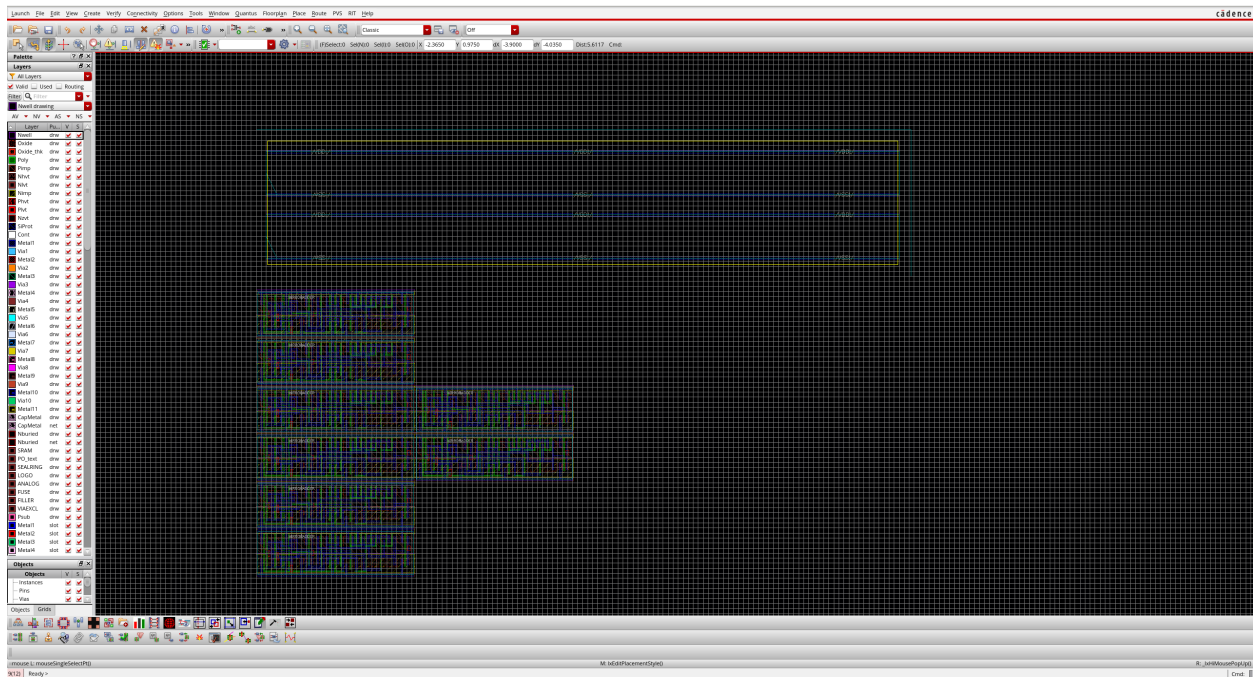


Figure 34: Placement Planning Complete, Once Standard Cell Row Planned

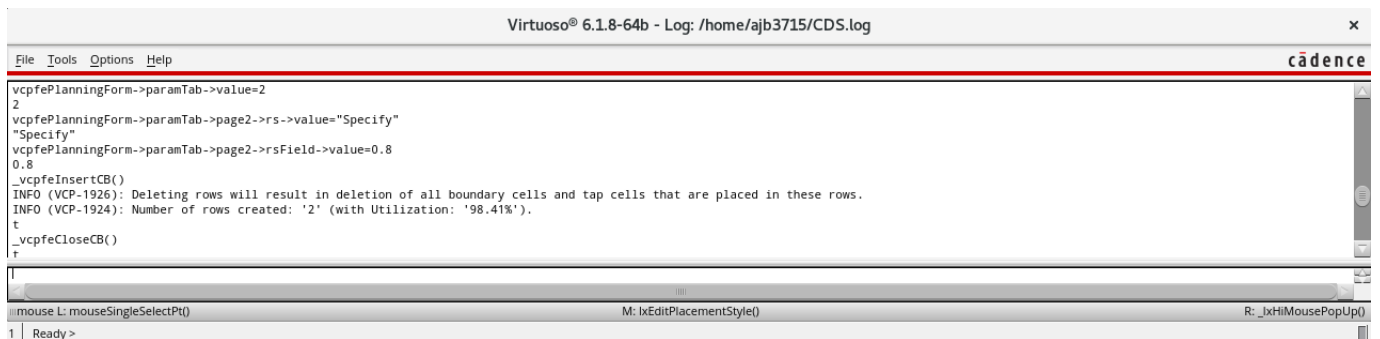


Figure 35: Placement Statistics Report in CIW

The tools can show you flight lines for incomplete nets (nets not yet routed); show hide all incomplete nets as shown in Figure 36.

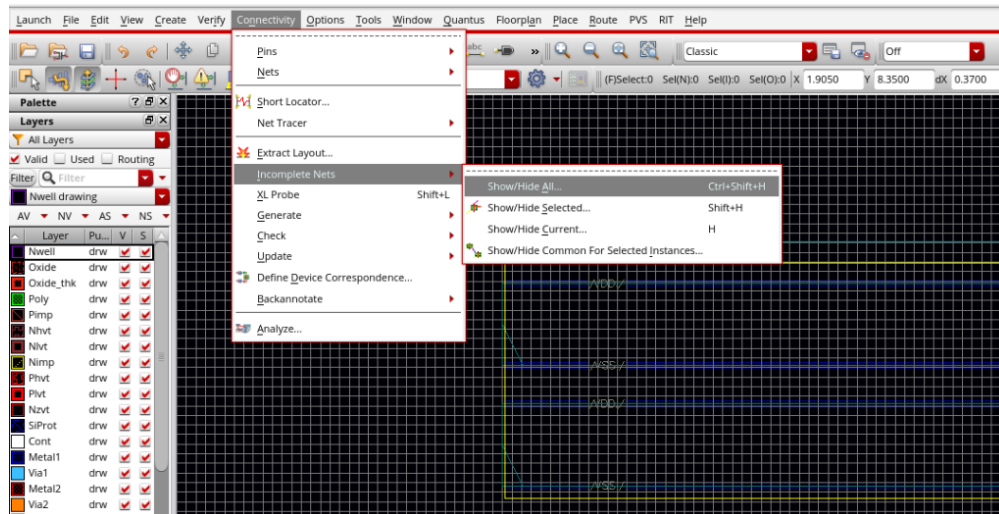


Figure 36: Launch Show / Hide All Incomplete Nets

All incomplete nets will be shown as in Figure 37 as highlighted flight lines.

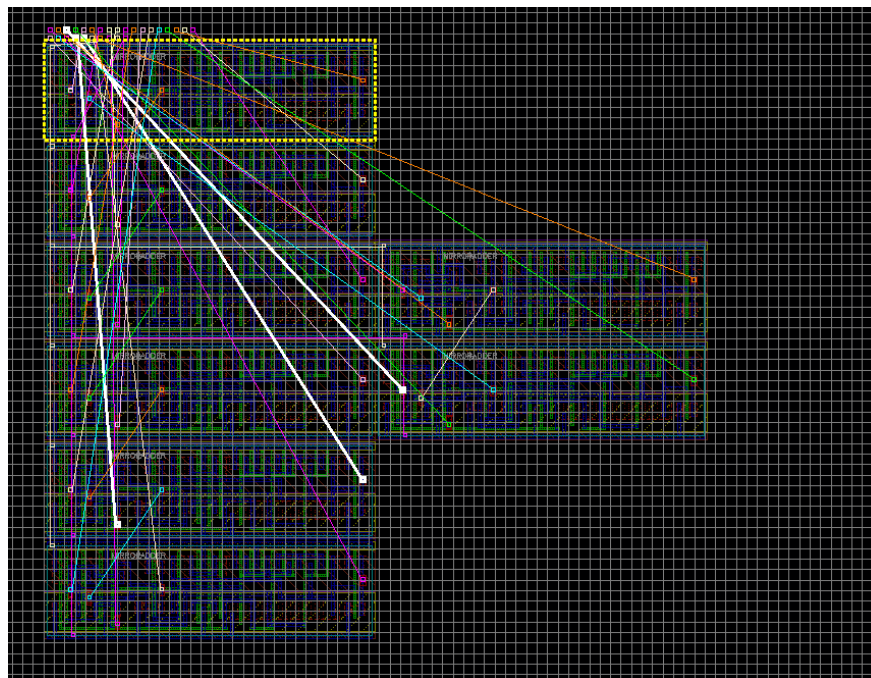


Figure 37: All Incomplete Nets Shown with Flight Lines

10 Place Design

To place the design we start by setting design parameters for the placer, select **Place** → **Custom Digital** → **Dressing Template Editor** as shown below.

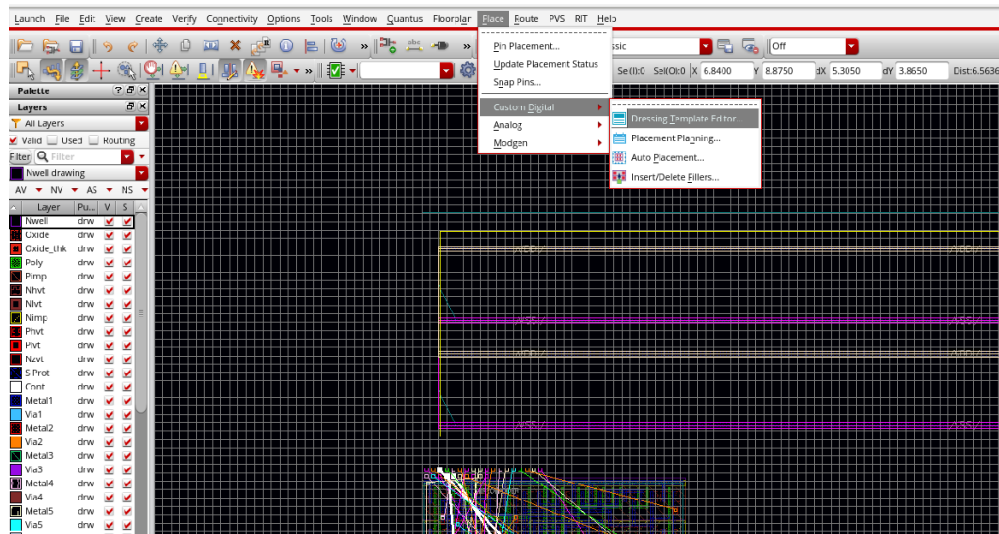


Figure 38: Launch Dressing Template Editor

Update the dressing template forms as shown in Figure 39, Figure 40 and Figure 41.

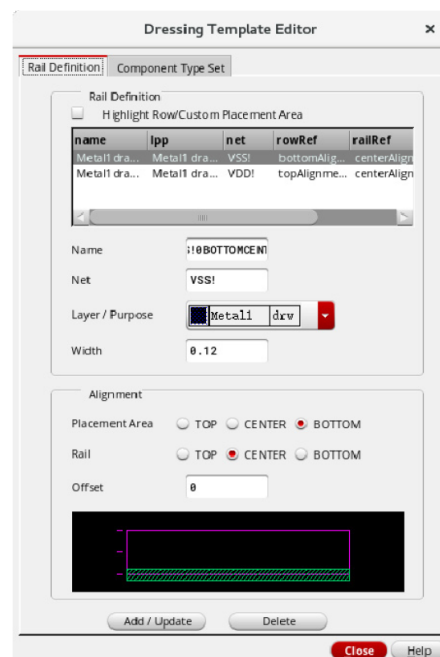


Figure 39: Dressing Template Editor - Rail Definition - VSS Rail

Click Add/Update to accept the settings, now adjust the VDD rail.

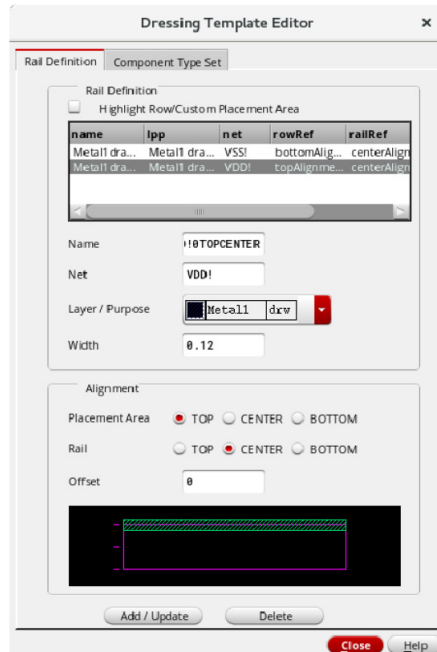


Figure 40: Dressing Template Editor - Rail Definition - VDD Rail

Click **Add / Update** to accept the settings , now adjust the **Component Type Set**.

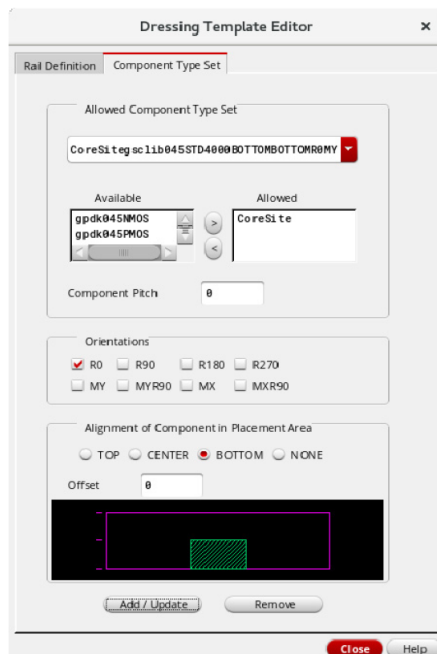


Figure 41: Dressing Template Editor - Component Type Set

Click **Add/Update** to accept the settings, and then Close the form.

We can now place the design; launch the digital placement tool, select Place → Custom Digital → Auto Placement as shown below (note that the image provides a preview of the final placed standard

cell row).

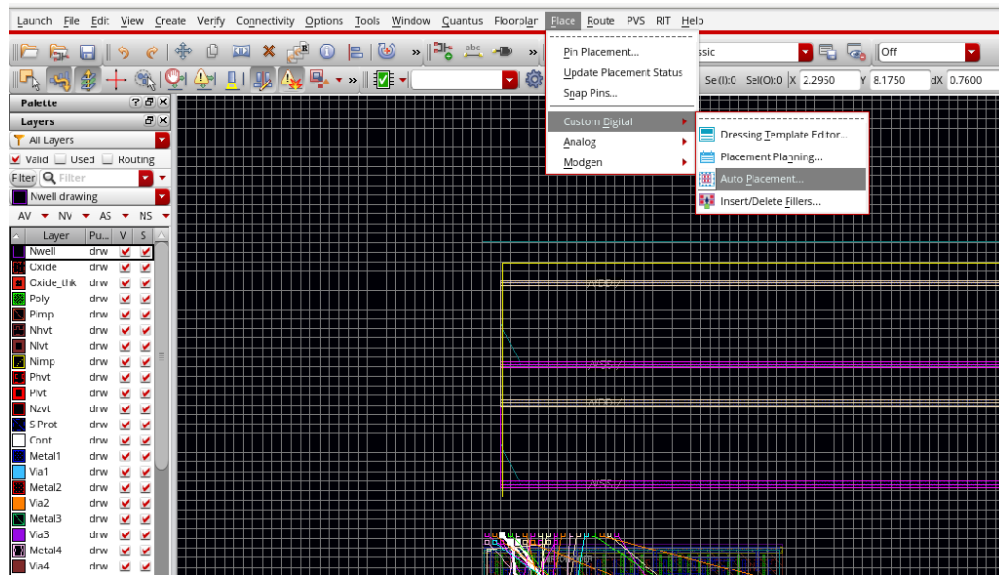


Figure 42: Launch Custom Digital Auto Placer. **Note that your layout will look different.**

Update the placement options form as shown.

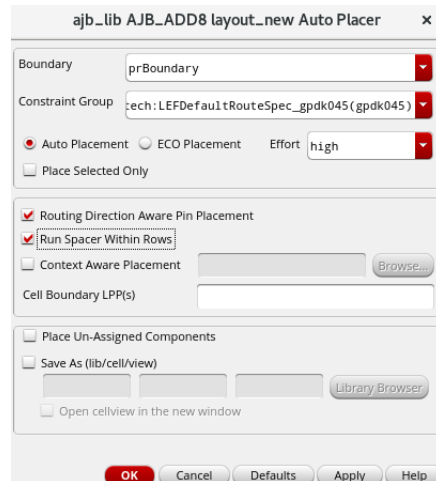


Figure 43: Launch Custom Digital Auto Placer Options

Click OK to place the design. Then, the design should be placed and occupy 100% of the target row as shown in Figure 44 below.

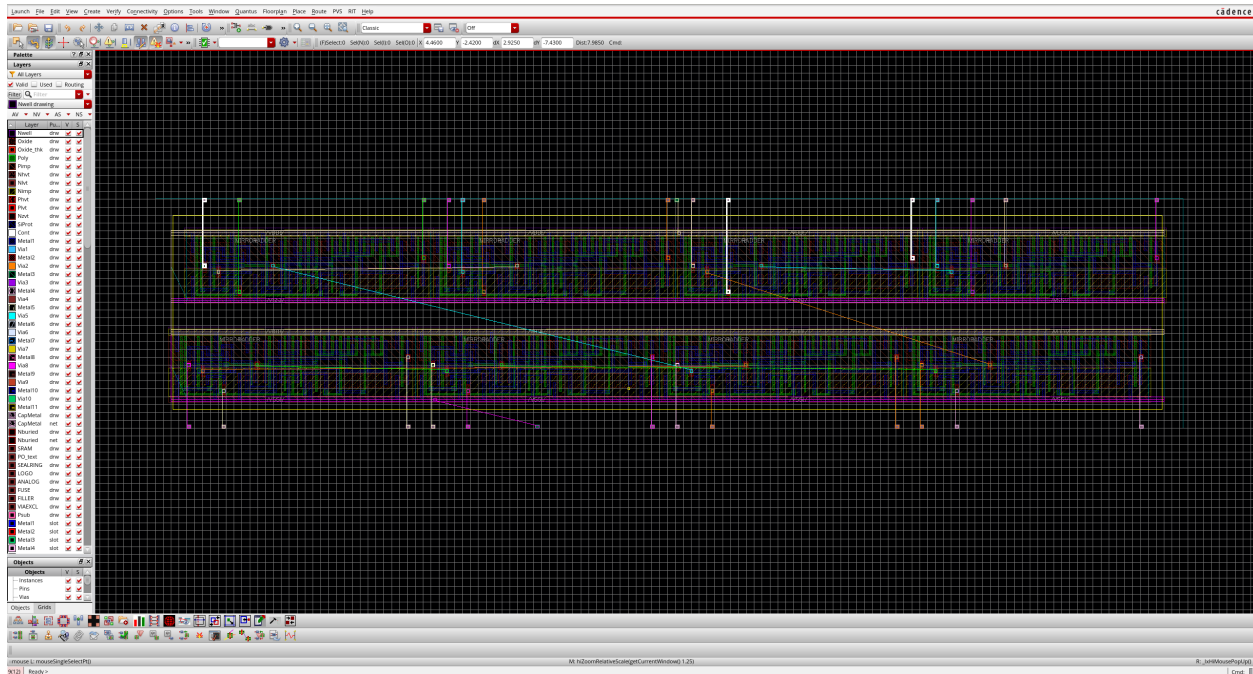


Figure 44: Placement Complete, Once Standard Cell Row Placed

Note that your pins will probably not be in these exact locations. we'll optimize the pin placement in the next step.

11 Optimize Pin Placement

I/O Pin placement is critical to optimize routing between cells and in a top down design we would probably define pin locations very early and enter the placement step generally with pins at fixed locations.

This is not always the case; assuming you do not know how your final routed cell will be used relative to other cells in your top level floorplan, you can use the following I/O Pin Placement rules as a guide:

1. Place all your inputs at the Top of your PR boundary with location Floating
 - Once placed, optimize pin placement
2. Place all your outputs at the Bottom of your PR boundary with location Floating.
 - Once placed, optimize pin placement
3. Place VSS on the one side of your PR boundary with location Floating.
 - Once placed, use the layout editor **Move** command to move the VSS pin and place it on top of one of the standard cell VSS rails. This should stop the router from routing the VSS bus.
4. Place VDD on the side opposite of VSS on your PR boundary with location Floating.

- Once placed, use the layout editor **Move** command to move the VDD Pin and place it on top of one of the standard cell VDD rails. This should stop the router from routing the VDD bus.
5. Even if you add the power straps prior to routing, you will need to place the VDD and VSS pins as directed.

For the purposes of this Tutorial, we have freedom with pin placement at this time, so with the cells placed, we can optimize the pin placement to minimize wire length. To launch the pin placement tool, select **Place** → **Pin Placement** as shown below (note that the image provides a preview of the final placed pins).

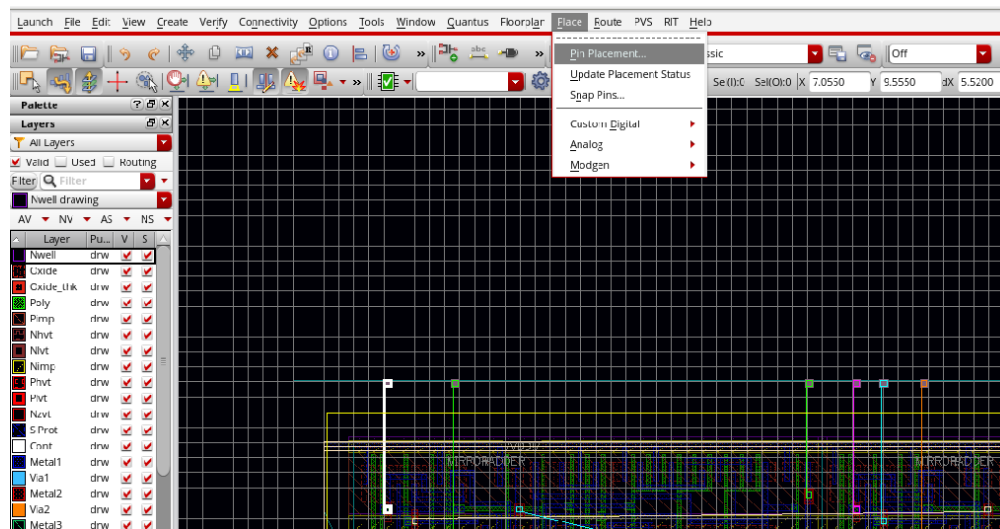


Figure 45: Launch Pin Placement

We'll start by configuring the various pin location (per edge) using this form.

The screenshot shows the 'Pin Placement: ajb_lib AJB_ADD8 layout_new' window. It has two tabs: 'Pin Planner' and 'Pin Optimization'. The 'Pin Optimization' tab is active.

Edit Section:

- ☒ Top Level Pins, ☐ Level-1 Pins
- ☐ Auto-Switch to Level-1
- Filter:** Edge (All), Status or Type (All), By Name (*.*)
- Place As In:** Schematic, Symbol
- Iterated Pins:** Interleave, Uninterleave, Expand, Collapse, ☐ All, ☒ Selected
- Change Order:** Move pins, Swap order
- Create:** VRail, HRail
- Zoom:** Zoom ON, 0, Min

Pin List Table:

Pin Name	Edge	Order	Status	Type	Layer	Width	Height
VDD!VDD!VDD!	any			IO	Metal3	0.080	0.080
VSS!VSS!VSS!	any			IO	Metal3	0.080	0.080
A1:A1:A1	any			I	Metal4	0.080	0.080
A2:A2:A2	any			I	Metal4	0.080	0.080
A3:A3:A3	any			I	Metal4	0.080	0.080
A4:A4:A4	any			I	Metal4	0.080	0.080
A5:A5:A5	any			I	Metal4	0.080	0.080
A6:A6:A6	any			I	Metal4	0.080	0.080
A7:A7:A7	any			I	Metal4	0.080	0.080
A8:A8:A8	any			I	Metal4	0.080	0.080
B1:B1:B1	any			I	Metal4	0.080	0.080
B2:B2:B2	any			I	Metal4	0.080	0.080
B3:B3:B3	any			I	Metal4	0.080	0.080
B4:B4:B4	any			I	Metal4	0.080	0.080
B5:B5:B5	any			I	Metal4	0.080	0.080

Message: Select Top Level Pin(s) from the list to edit attributes.

Attributes:

- Edge: Any
- Order:
- Placement Status Constraint: No Constraint (Floating)
- Apply

Shape:

- Layer: Metal3 pin
- Width: 0.080
- Height: 0.080
- Re-Place Pins: All Edges

Pin Spacing:

- Value:
- Spacing type: center-center
- Update Constraints: ☒
- Space from LEFT/BOTTOM pin
- Space from CENTER pin
- Space from RIGHT/TOP pin

Buttons: Close, Defaults, Help

Figure 46: Pin Placement Options Form

Adjust the pin locations as follows: (i) Input Pins on top of the edge; (ii) Output pins are placed on the bottom edge; (iii) VDD/VSS power Pins are placed on the left edge. Select the pins you wish to change in the listing area (all the output signals pins are selected) as shown in Figure 47 below. We will also adjust the Metal layer for the VDD and VSS pins, as this will aid us with power strapping later.

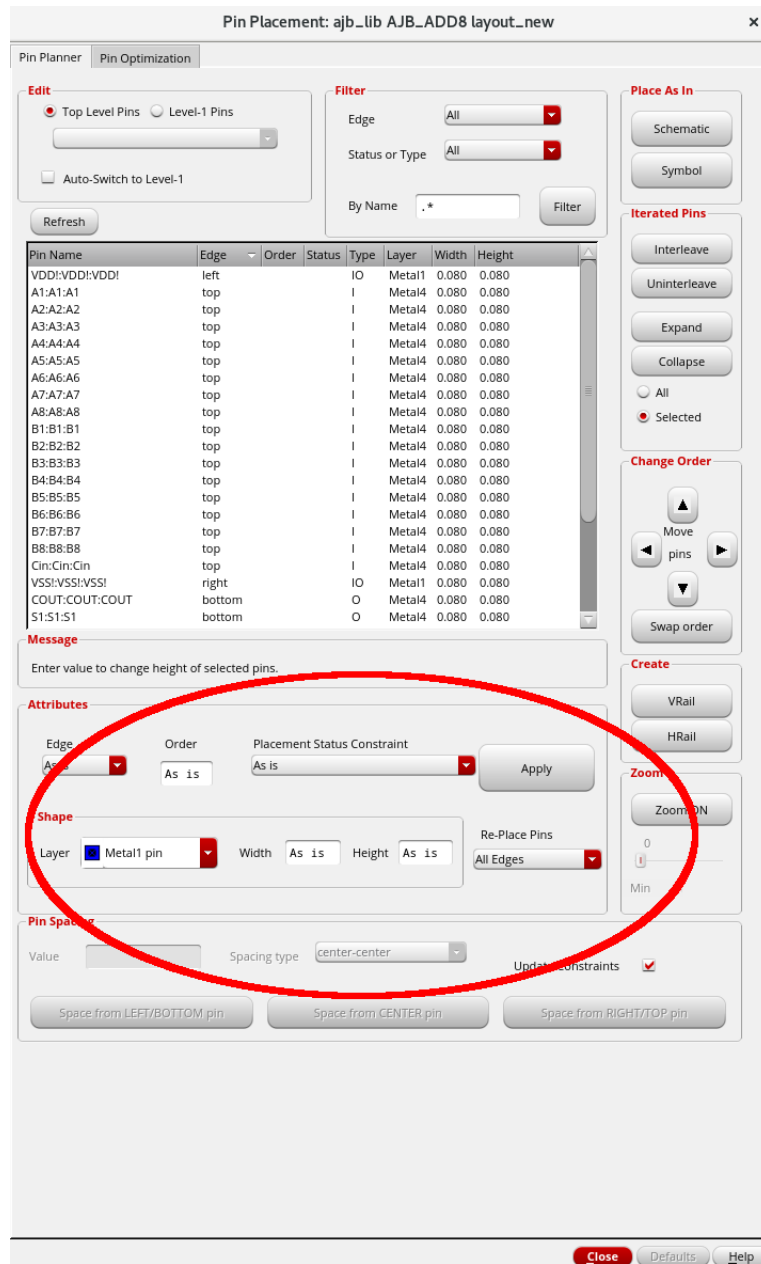
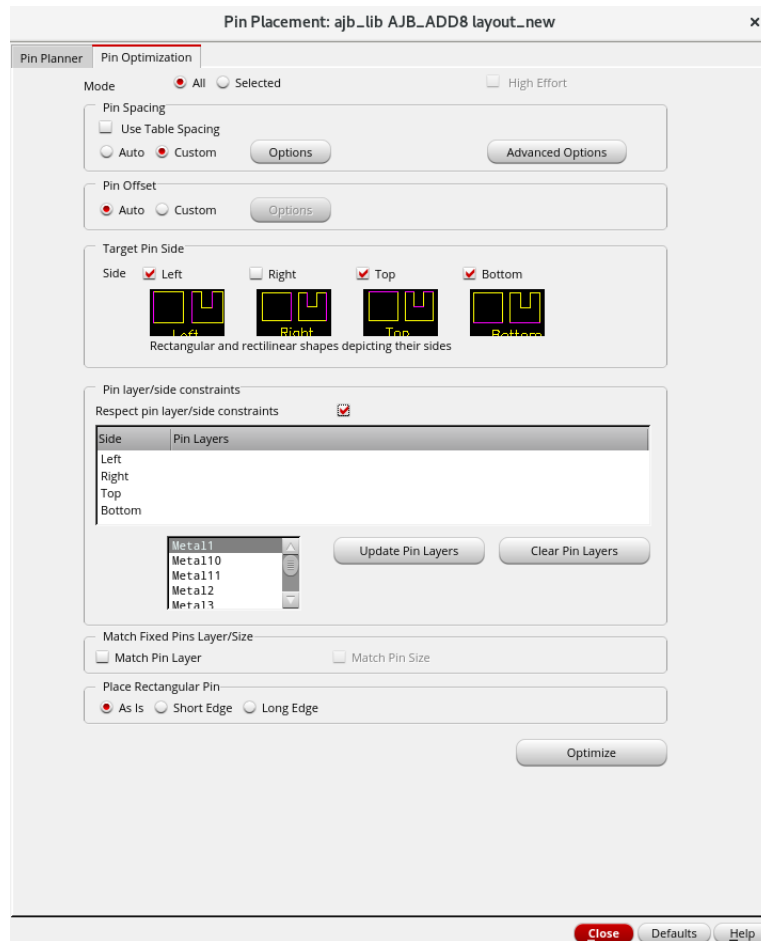


Figure 47: Pin Placement Form - Assigning Edges

In the attributes area, configure the Edge - for the output pins select **Bottom**, **Placement Status Constraint** - select **No Constraint (Floating)**; then click Apply to update the location of the selected pins.

At this point you should observe the incomplete net flight lines; perhaps you may notice that the pin placement is not optimal based on the length and direction of the flight lines.

For a more optimal pin placement solution, you can use the pin optimization engine to 'optimize' the pin placement by selecting the **Pin Optimization** tab. We'll setup some spacing options.



Pin Placement: ajb_lib AJB_ADD8 layout_new

Pin Planner Pin Optimization

Mode ☒ All ☐ Selected ☐ High Effort

Pin Spacing ☐ Use Table Spacing ☐ Auto ☒ Custom Options Advanced Options

Pin Offset ☒ Auto ☐ Custom Options

Target Pin Side Side ☒ Left ☐ Right ☒ Top ☒ Bottom

Rectangular and rectilinear shapes depicting their sides

Pin layer/side constraints Respect pin layer/side constraints ☒

Side	Pin Layers
Left	
Right	
Top	
Bottom	

Update Pin Layers Clear Pin Layers

Match Fixed Pins Layer/Size ☐ Match Pin Layer ☐ Match Pin Size

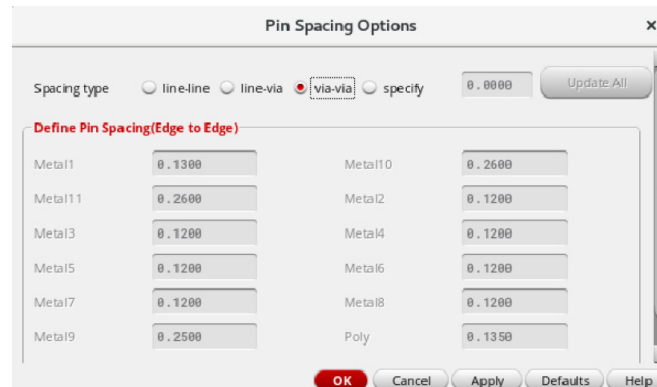
Place Rectangular Pin ☒ As Is ☐ Short Edge ☐ Long Edge

Optimize

Close Defaults Help

Figure 48: Pin Placement Optimization Form

Select **Pin Spacing** → **Custom** and click **Options**.



Pin Spacing Options

Spacing type ☐ line-line ☐ line-via ☒ via-via ☐ specify 0.0000 Update All

Define Pin Spacing(Edge to Edge)

Metal1	0.1300	Metal10	0.2600
Metal11	0.2600	Metal12	0.1200
Metal13	0.1200	Metal14	0.1200
Metal15	0.1200	Metal16	0.1200
Metal17	0.1200	Metal18	0.1200
Metal19	0.2500	Poly	0.1350

OK Cancel Apply Defaults Help

Figure 49: Pin Optimization - Pin Spacing Options Form

Update the form as shown in click OK. You can allow the tool to optimize placement on one or more sides by selecting the region(s) as shown in Figure 48 above. Click the **Optimize** button to run the pin optimization engine for the region(s) selected. Now, your pins and placed design should

appear as shown below.

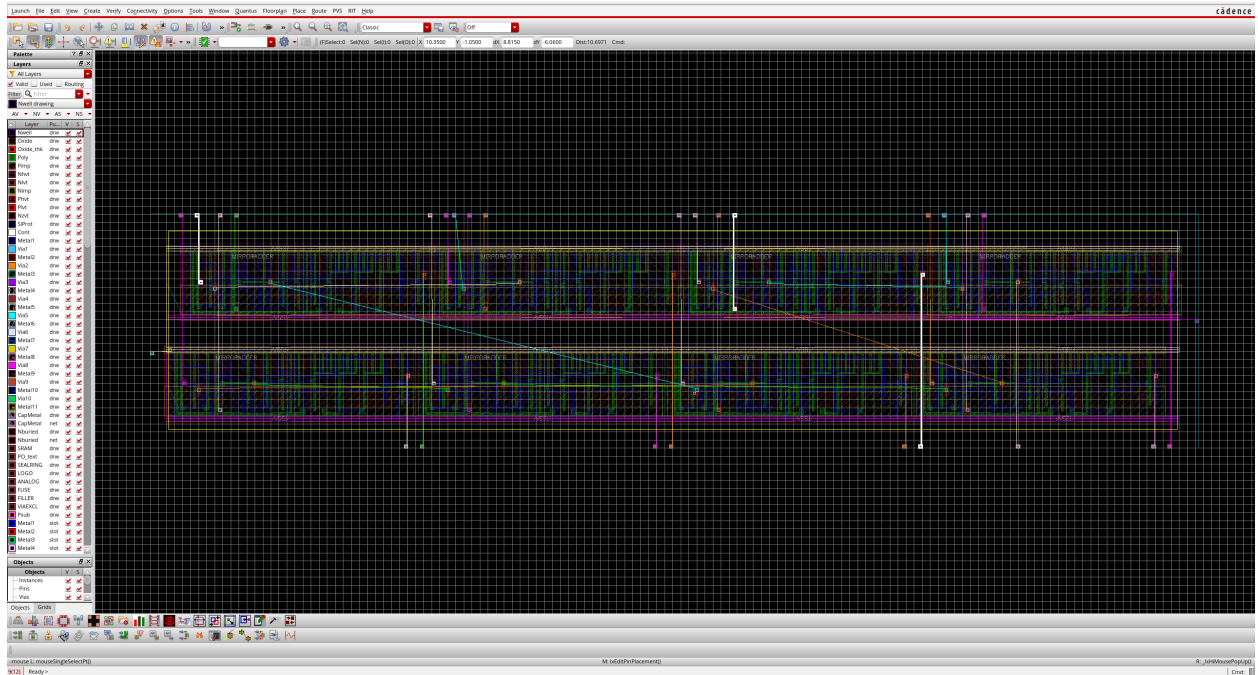


Figure 50: Pin Optimization Complete (with final routing showing too)

Note that this is a very simple example; more complex blocks, and block to block wiring will have a direct impact on pin and cell placement within a block.

12 Power Strapping

Power strapping is a method used to carry VDD and VSS signals throughout multiple rows within the design, and can be a useful tool later when attempting to integrate the adder into another circuit (Like an ALU for instance).

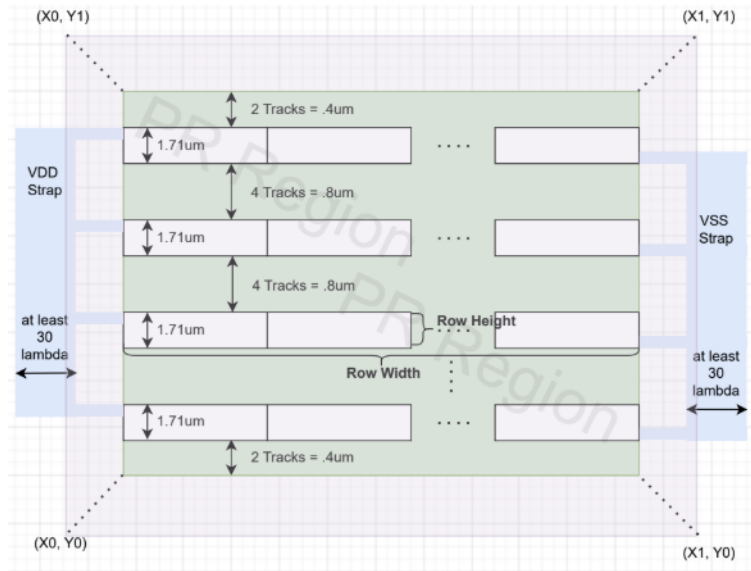


Figure 51: Power Strapping Setup)

Note that VDD and VSS are not mandated to be on the left or right sides of the design. Using the Metal 1 draw tool, draw the VDD and VSS straps on the edge of the design as seen in Figures 52 and 53.

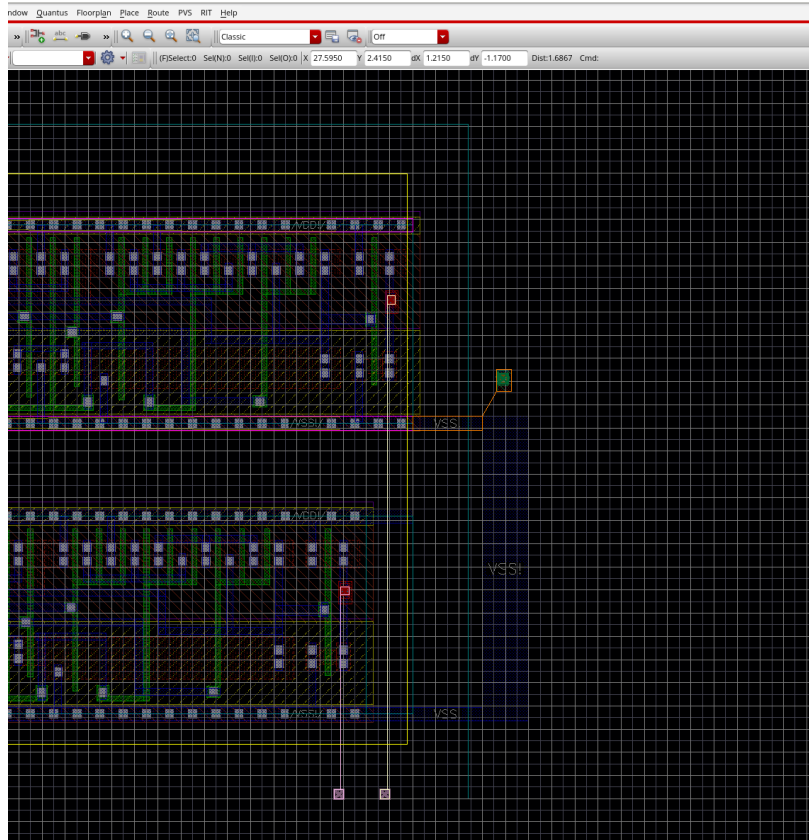


Figure 53: Power Strapping Setup - VSS

Now, align the VDD and VSS pins overtop of their respective powerstraps and connect them to the powerstraps as shown in Figure 54.

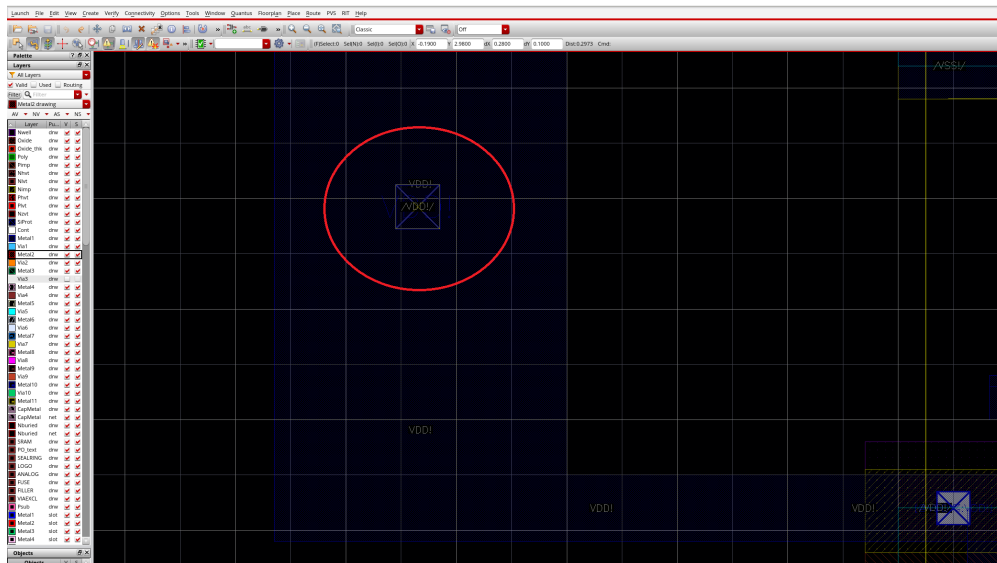


Figure 54: Power Strapping Setup - Pins

13 Wire Planning

As with most tools, there are a number of ways to access the Virtuoso Space-Based Mixed0Signal Router (VSR) integrated into Virtuoso GXL. We will use the Wire Planning assistant; select **Window** → **Assistants** → **Wire Assistant** as shown below.

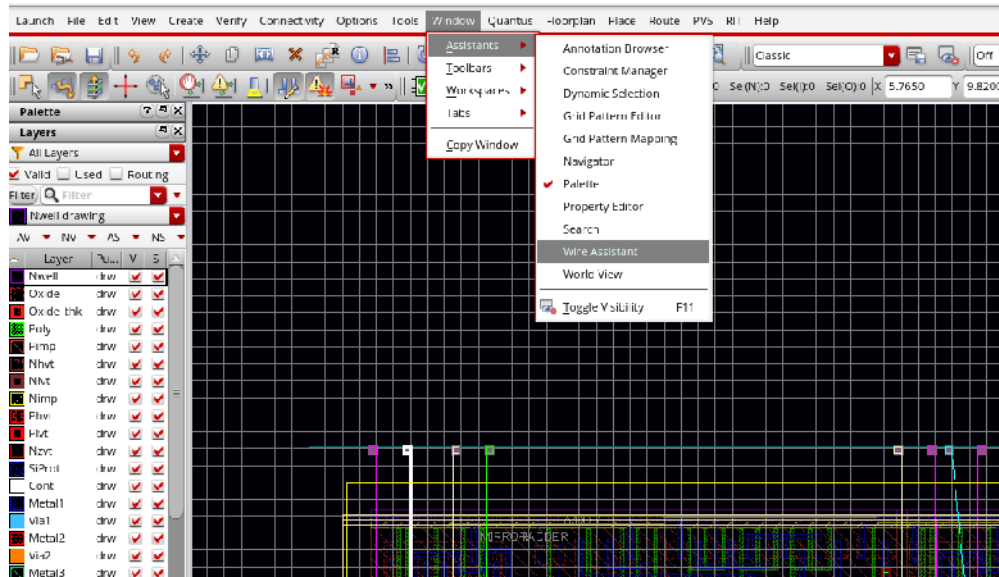


Figure 55: Wire Assistant

WARNING: VSR does not currently take timing or power constraints into consideration when routing.

A new Wire Assistant panel will be added to your layout window as shown below.

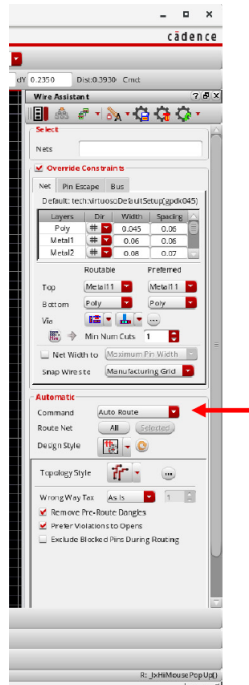


Figure 56: Wire Assistant Panel

WARNING: Be aware that changing the Command at any time MAY clear any previously set options (meaning you will **lose** any settings you modified - always inspect the settings before starting **Auto Route**).

OPTIONAL: You can click the Floppy Disk icon near the top of the Wire Assistant form to save your Wire Assistant settings - click on the Disk and a form will appear. Choose a name for your saved settings and close the form.

We'll start by seeding the various forms by clicking on the constraint icon, and selecting a base constrain group as shown below.

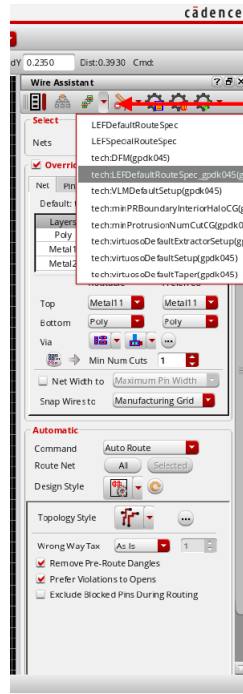


Figure 57: Wire Assistant - Select Base Constraint Group

Now that we've seeded the settings, we'll configure the various parameters we need to effective routing we're only going to make changes to the **Net** tab as shown below.

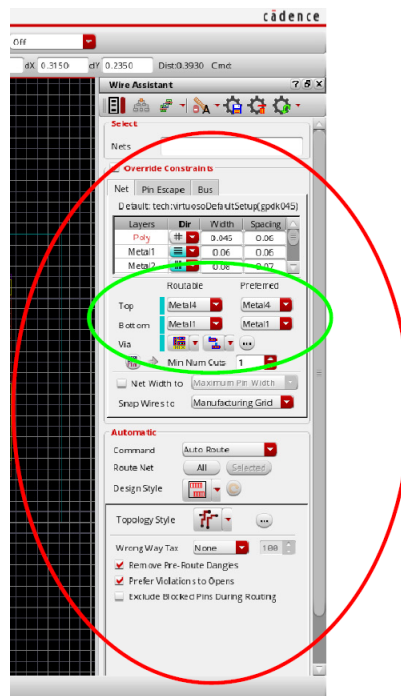


Figure 58: Wire Assistant - Route Planning Setup

To start we will only be allowing the router to route using Metal 4 → Metal 1. We'll use the **Top**

and **Bottom** radio buttons as shown in Figure 58 to select Metal 4 as the Top layer, and Metal 1 as the bottom layer.

Next using the Layers, which are just above the Top and Bottom radio buttons, we'll set up the allowed routing direction for the layers used. Note that Poly has a red highlight since this layer cannot be used for routing. The layers have a radio button with 3 selections: (i) meaning routing is allowed in any direction; (ii) meaning that routing is restricted to the vertical direction, and (iii) allows both.

Consistent with the rules contained in the Generic 45nm standard cell technology library README file, setup the routing direction restrictions as follows: (i) Metal 1, Horizontal; (ii) Metal 2 Vertical; (iii) Metal 3, Horizontal; (iv) Metal 4, Vertical.

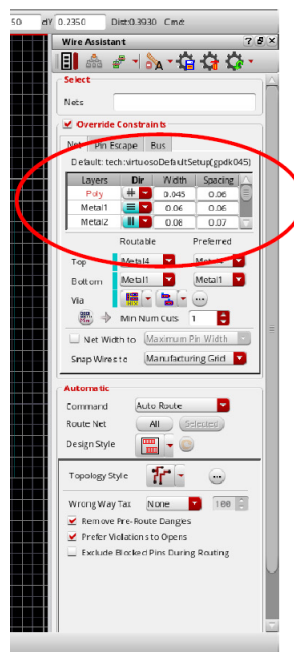


Figure 59: Wire Assistant - Route Planning Setup

In the Via area highlighted below, we'll make the changes to how the router creates vias.

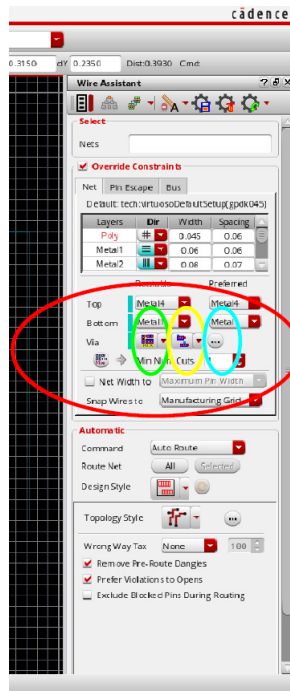


Figure 60: Wire Assistant - Route Planning Setup

The first via option (green oval) selects the type of via created, select the option **Automatic Enclosure and Cut Direction**

The second Via option (yellow oval) selects how wiring will enter a via, selected the option Offset or Center to the Wire.

The third Via option (blue oval) selects the layer based via configuration, click on the "...” and a Via Configurations window will open as shown below.

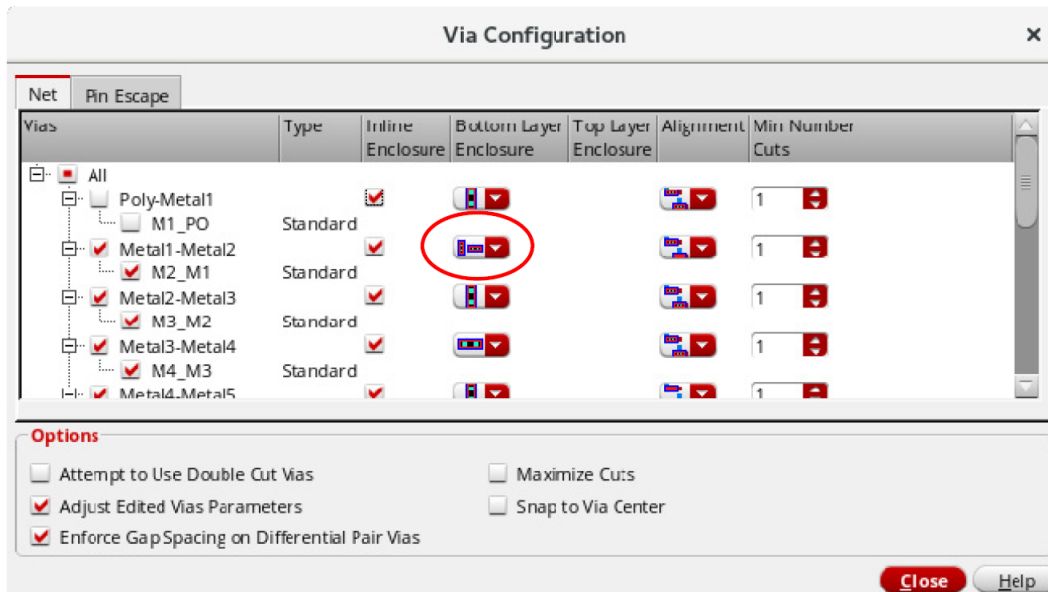


Figure 61: Wire Assistant - Via Configuration Setup

Under the Net tab change the Metal1-Metal2 via Bottom Layer Enclosure to allow the enclosure to be created in either routing direction as highlighted above. We will not be making any changes to the Pin Escape tab. Now, click Close to accept the Via Configuration

As we work our way down the form, make sure the Snap Wired to setting is Manufacturing Grid.

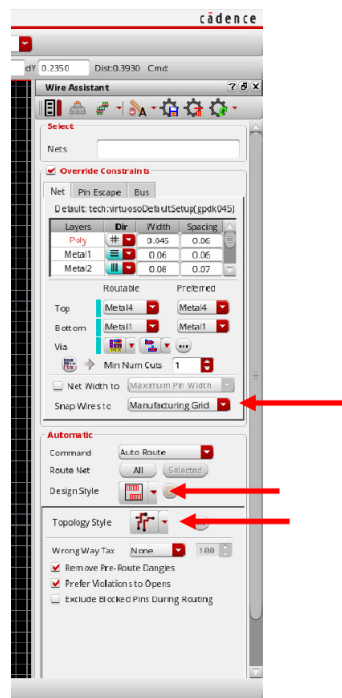


Figure 62: Wire Assistant - Route Planning Setup

The two icons radio buttons below Snap Wires to configure the routing algorithm and strategy used by the router.

The **Design Style** radio button selects the Routing Flow used, select ASIC (this configures how the various routers, such as global, local, detail, etc. are utilized during routing).

The **Topology Style** radio button selects the Routing Topology used, select **Minimum Spanning Tree** (the goal of this routing topology is to minimize wire length).

The final setting is to define how the defined layer direction is honored by the router.

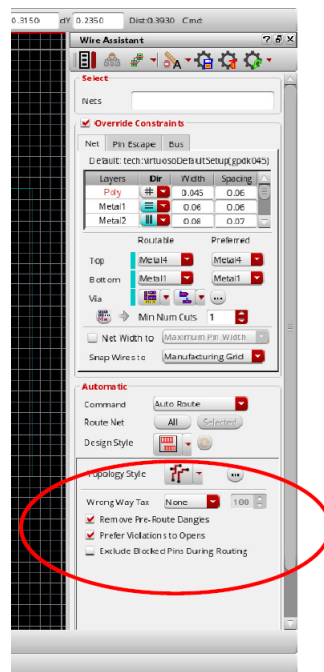


Figure 63: Wire Assistant - Route Planning Setup

Using the **Wrong Way Tax** radio button, select **None** to restrict the routing from routing in the opposite direction by setting cost penalty for misusing a metal layer.

Finally make sure the options **Remove Pre-Route Dangles** and **Prefer Violations to Opens** are both checked.

You are now ready to route your design!

14 Routing

Assuming all the required settings have been made, your Wire Assistant form should look like the one below.

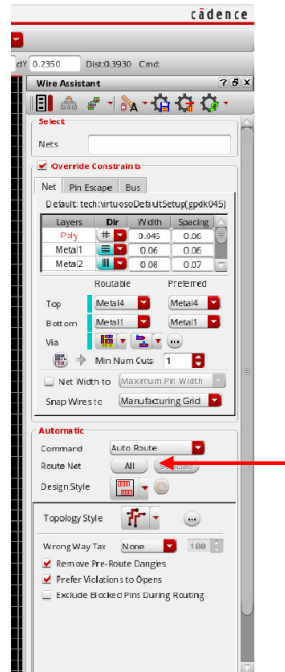


Figure 64: Wire Assistant - Automatic Routing

Click on the **Route Net All** button to start the router. Now, your design will be automatically routed, and the router log messages will be displayed in the CIW as shown below.

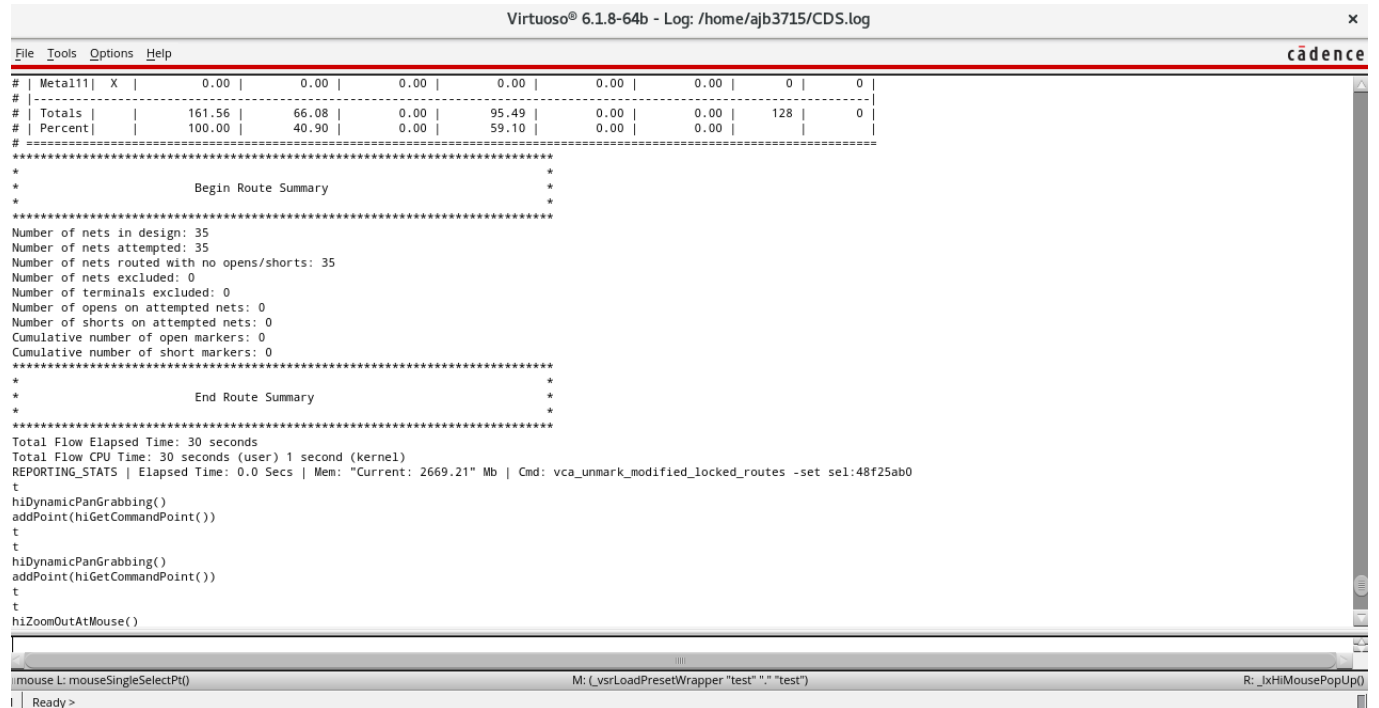


Figure 65: Wire Assistant - Router Log Messages in CIW

while there are a lot of messages logged, the Route Summary as circled above will provide details

on how much of your design was successfully routed. For this one, we know it was 100% routed. The final routed design should appear similar to Figure 65 below.

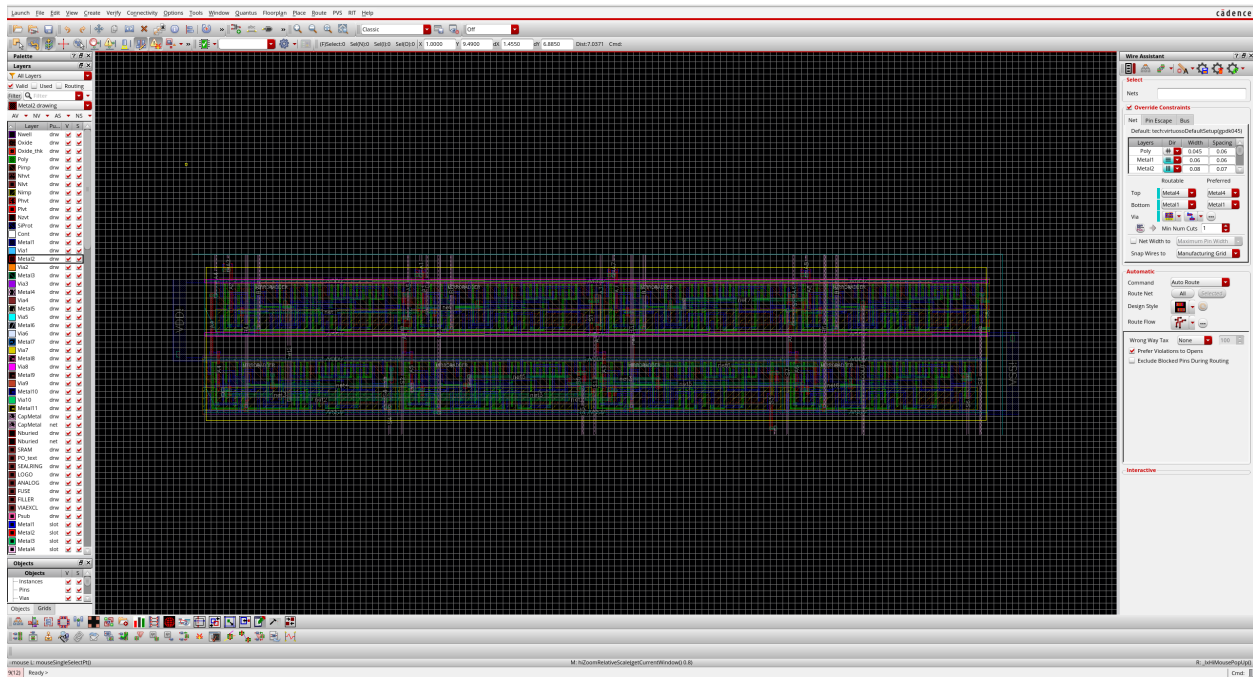


Figure 66: Final Fully Routed Layout

Check the routing report in the CIW.; If the final routed design has any shorts or opens, you can run the **Command Fix Violations**; depending on how clean the route appears you can also run the **Command Optimization**. Note that you may need to run the Auto Route, Fix Violations, and optionally Optimization commands multiple times to obtain an ideal route.

WARNING: Be aware that changing the Command at any time MAY clear any previously set options (meaning you will **lose** any settings you modified - always inspect the settings before starting **Auto Route**).

OPTIONAL: You can click the Floppy Disk icon near the top of the Wire Assistant form to save your Wire Assistant settings - click on the Disk and a form will appear. Choose a name for your saved settings and close the form.

15 DRC

Congratulations, your design was 100% routed! How good of a job did the router do? We don't actually know yet but DRC will let us know. Select **PVS** → **DRC** and fill out the forms as previously done in other exercises. Assuming you set things up correctly prior to routing, your layout should be DRC clean as shown below.

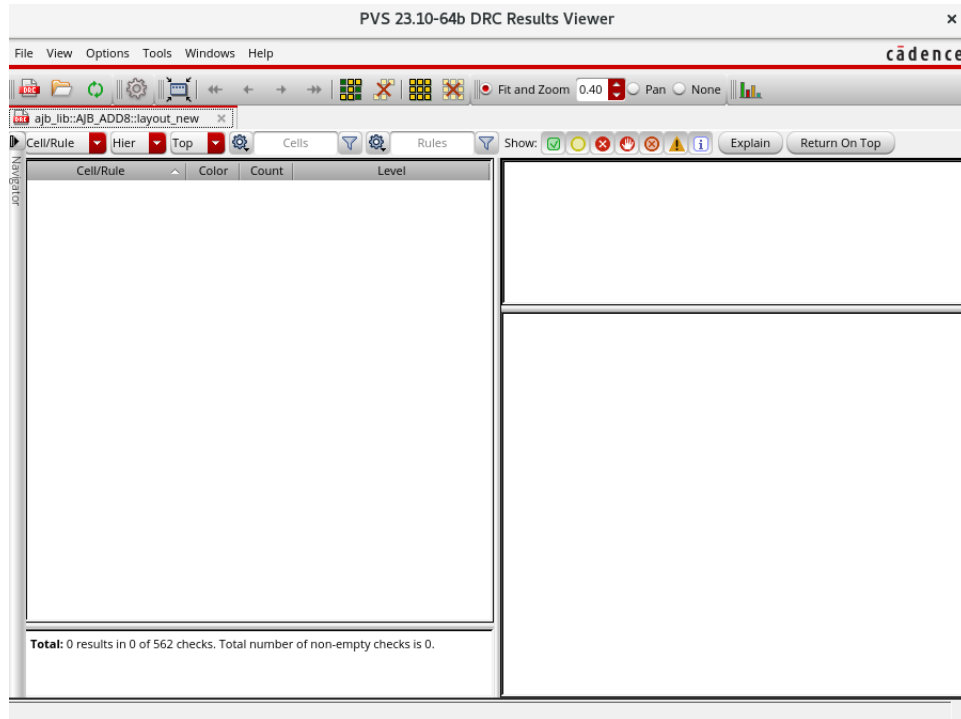


Figure 67: PVS DRC Log Window

16 LVS

Next run LVS ,Select **PVS** → **LVS** and fill out the forms as previously done in other exercises. Assuming your design was wired and labeled correctly, your layout should be LVS clean as shown below.

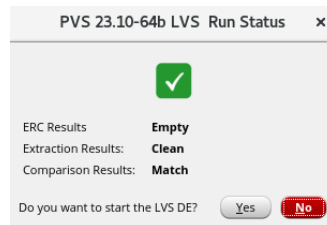


Figure 68: PVS LVS Results Window

17 Run Parasitic Resistor, Capacitor(RC) Extraction (RCX)

Assuming your LVS successfully passes without mismatched, **QRC** → **Run PVS-Quantus QRC** and fill out the forms as previously done in other exercises. Simulating with extracted data would allow you to determine the impact of routing on the timing performance of your block.

Congratulations, you've successfully completed your first block place and route! This is the end of Part 1.

18 Part 2: Analog — Digital Mixed-Signal Simulation

While spice level analog simulation is important for the circuit design and characterization for standard cells, as standard cells are combined into larger and larger blocks, the capacity of analog simulators limits a designer's ability to complete simulations in reasonable amounts of time. There is also the need to validate the operation for analog and digital circuits that are wired together. Analog— digital mixed-signal simulation bridges the gap between the two primary circuit design domains and provides a powerful mechanism to the design exploration and validation space.

To start we'll create a testbench using Verilog HDL to drive your FML_ADD8 cell. In the Library Manager window, select the library you created in Lab Ex 1, then select **File** → **New** → **Cellview**. Create the view functional and give your cell functional the name FML_ADD8_TESTBENCH (again, using your initials).

```
1 //Verilog HDL for "ajb_lib", "AJB_ADD8_TESTBENCH" "functional"
2
3 module AJB_ADD8_TESTBENCH (A1, A2, A3, A4, A5, A6, A7, A8, B1, B2, B3, B4, B5, B6, B7, B8, S1, S2, S3, S4, S5, S6, S7, S8, Cout, Cin );
4 input S1, S2, S3, S4, S5, S6, S7, S8, Cout;
5 output A1, A2, A3, A4, A5, A6, A7, A8, B1, B2, B3, B4, B5, B6, B7, B8, Cin;
6
7 wire A1, A2, A3, A4, A5, A6, A7, A8;
8 wire B1, B2, B3, B4, B5, B6, B7, B8;
9 reg [8:1]A = 0;
10 reg [8:1]B = 0;
11 reg Cin;
12 wire [8:1] sum = {S8, S7, S6, S5, S4, S3, S2, S1};
13 integer i;
14 integer passed;
15 assign {A8, A7, A6, A5, A4, A3, A2, A1} = A;
16 assign {B8, B7, B6, B5, B4, B3, B2, B1} = B;
17
18 //assign Cout = R9;
19
20 initial
21 begin
22     passed = 0;
23     for (i = 0; i < 500; i = i + 1) begin
24         A = $random();
25         B = $random();
26         Cin = $random() %2;
27
28         $display("Round %d A = %b, B = %b, Cin = %d" , i, A, B, Cin);
29
30         #10;
31
32         $display(" Sum = %b, Cout = %b", sum, Cout);
33
34         #10;
35
36         if((sum + (Cout << 8)) != A + B + Cin) begin
37             $display(" It failed ");
38             passed = passed -1;
39         end
40
41         passed = passed + 1;
42     end
43
44     $display("Passed tests = %d", passed);
45
46     #10;
47
48     A = 255;
49     B = 255;
50     Cin = 1;
51
52     $display("Round %d A = %b, B = %b, Cin = %d" , i, A, B, Cin);
53
54     #10;
55
56     $display(" Sum = %b, Cout = %b", sum, Cout);
57
58     #10;
59
60     if((sum + (Cout << 8)) != A + B + Cin) begin
61         $display(" It failed ");
62     end
63 end
64
65
66 end
67 endmodule
68
69
```

Figure 69: FML_ADD8.TESTBENCH Functional View

Check and save your Verilog HDL. Assuming the code is syntax error free (check for errors in the CIW), use the menu pick **Create** → **Cellview From Cellview** to create a symbol view for FML_ADD8.TESTBENCH.

Now we'll create a schematic to wire your new testbench, FML_ADD8.TESTBENCH, to your FML_ADD8 cell. IN the library Manager window, select your library and then select **File** → **New** → **Cellview**. Create the view schematic and give your cell schematic the name FML_ADD8.TEST. Add labels to all the connections as seen in the Figure Below.

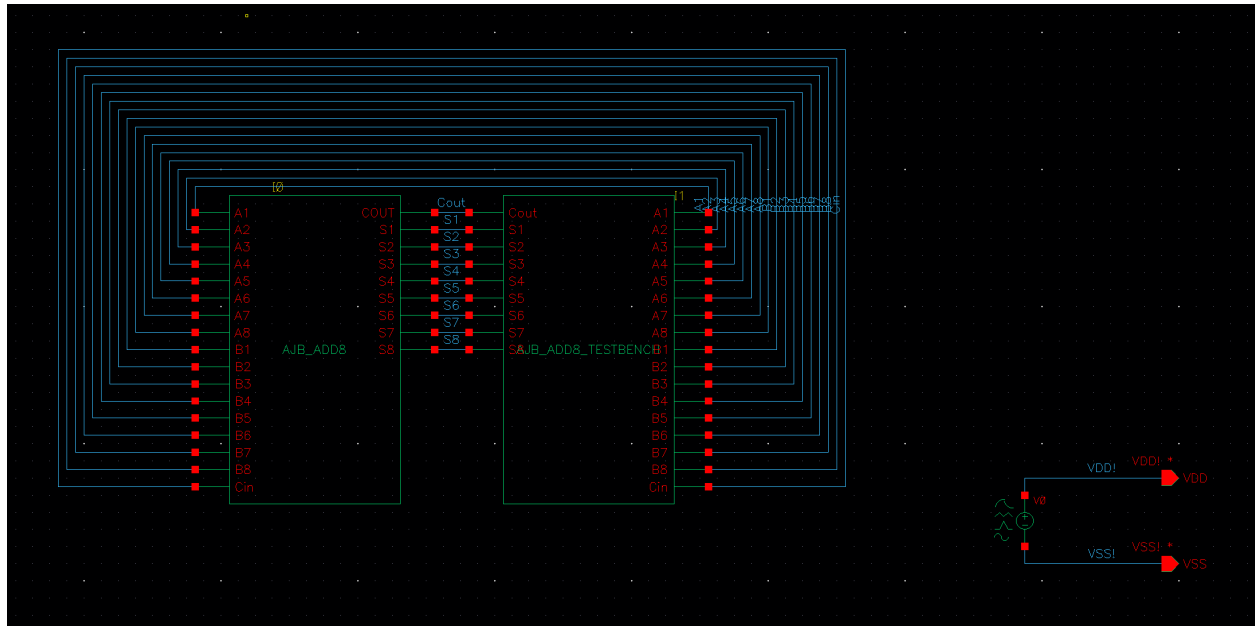


Figure 70: FML_ADD8.TESTBENCH Schematic View

For convenience, we'll use a supply cell, vsourse, from the analogLib to power the test schematic as shown in Figure 71 - note the **source type** is DC.

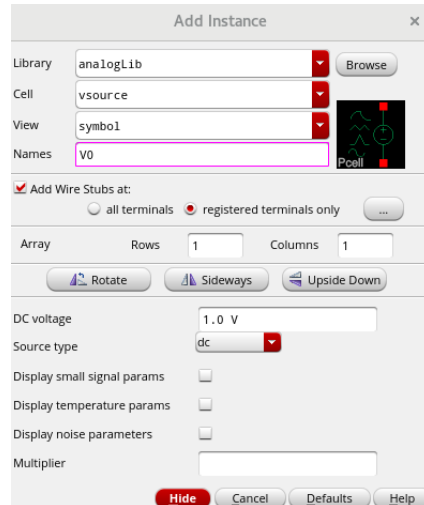


Figure 71: Vsource Configuration

With all the necessary components in your library, let's start the AMS process. The first step is to create a simulation configuration. In the Library Manager window, select your library and then select **File** → **New** → **Cellview**. Create the view config for your cell FML_ADD8_TEST.



Figure 72: Create cell FML_ADD8_TEST View config

A new configuration setup will appear. Now setup the configuration options; click on the **Library List** setup button (“...”).

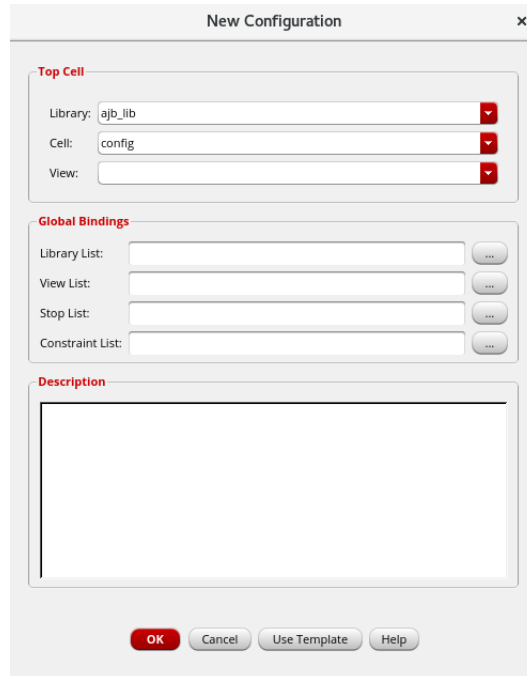


Figure 73: Create cell FML_ADD8_TEST View configuration Setup

Add the libraries in the order shown in Figure 74.

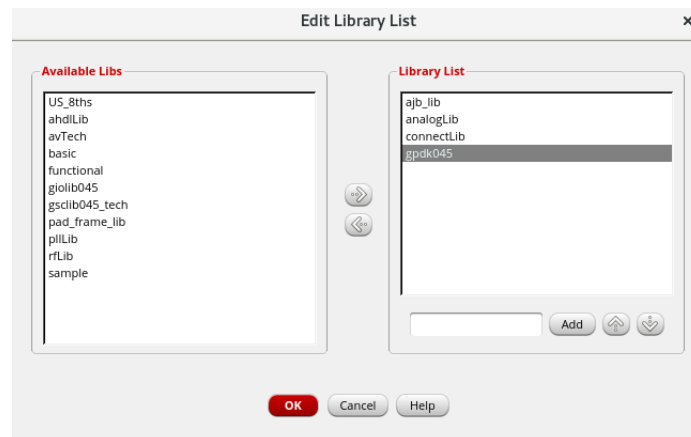


Figure 74: Edit Library List

Click OK to continue and you will return to the new configuration form. ON the new configuration options window; click on the View List setup button (“...”).

Add the views in the order shown in Figure 75.

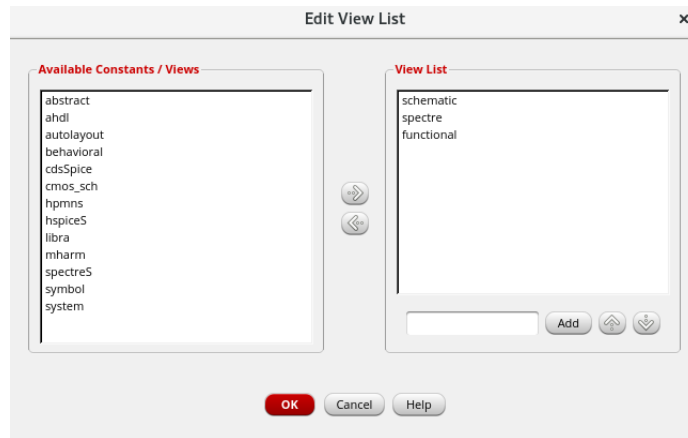


Figure 75: Edit View List

Click OK to continue. Now on the new configuration options window, click on the Stop List setup button (“...”).

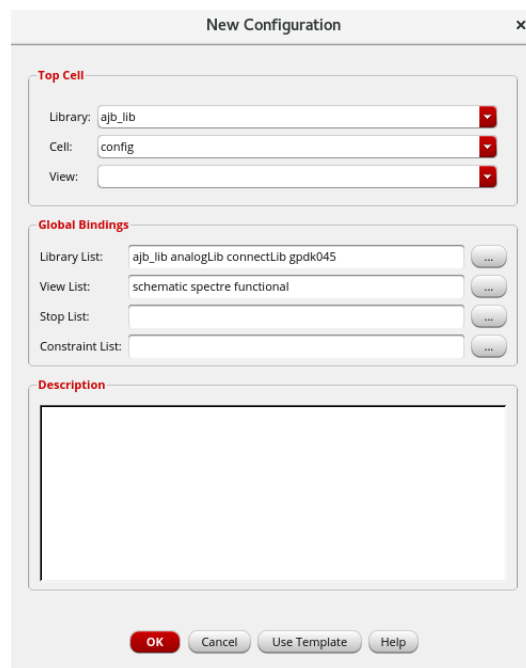


Figure 76: Cell FML_ADD8_TEST View configuration Final Setup

Click OK to continue. Then, add the views in order shown in Figure 78.

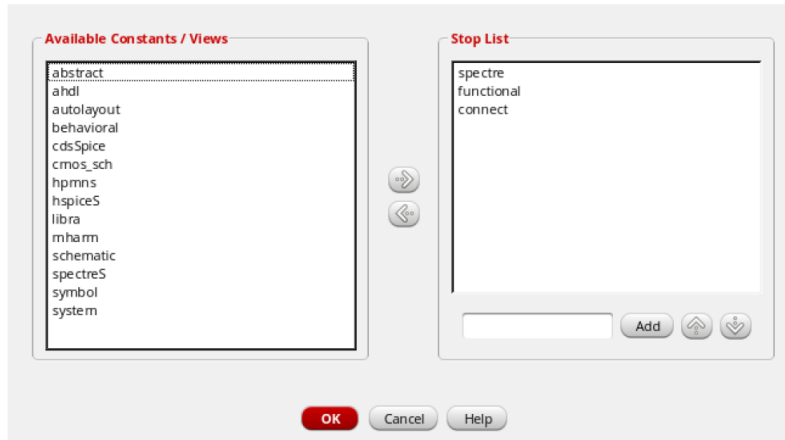


Figure 77: Edit Stop List

Click OK to continue. Finally OK the **New Configuration** form.



Figure 78: Final Configuration

The hierarchy editor will open. The config view is used by the simulation netlister to control how the design is represented in the generated netlist. For example in the config view shown in Figure 79, the “top” cell, FML_ADD8_TEST, is represented by a schematic.

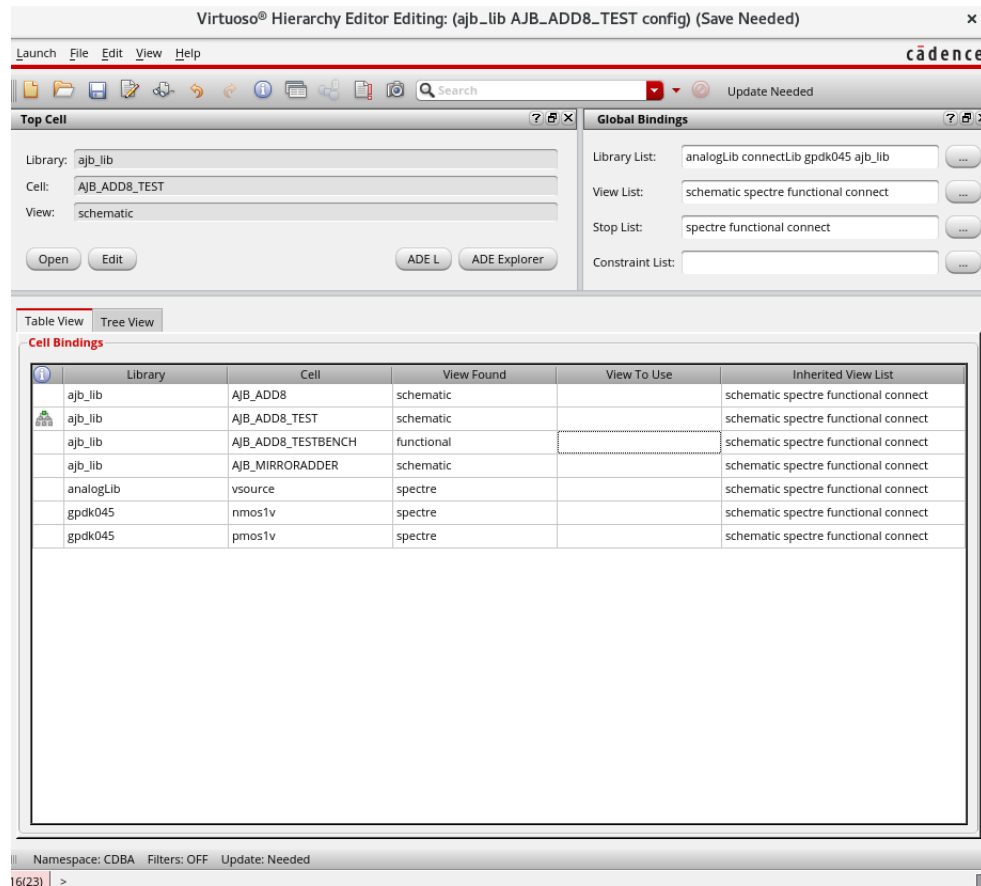


Figure 79: Cell FML_ADD8_TEST View Config Hierarchy Editor

As necessary, you can “override” the view to use on a component by component basis by entering the required view in the **View To Use** column entries. Update the hierarchy to override the views as shown in Figure 80; click the **Reconfigure Hierarchy** button to update the hierarchy.

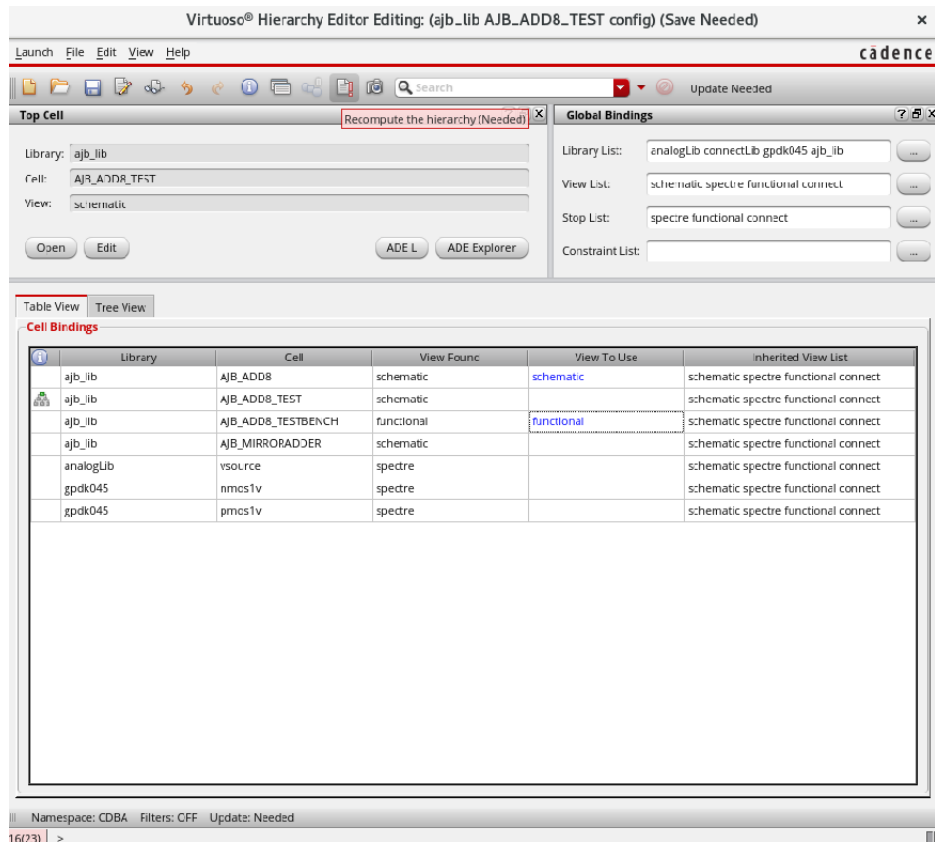


Figure 80: Cell FML_ADD8.TEST View Config Hierarchy Editor Overrides

At this point the hierarchy editor and schematic windows should be open as shown in Figure 81.

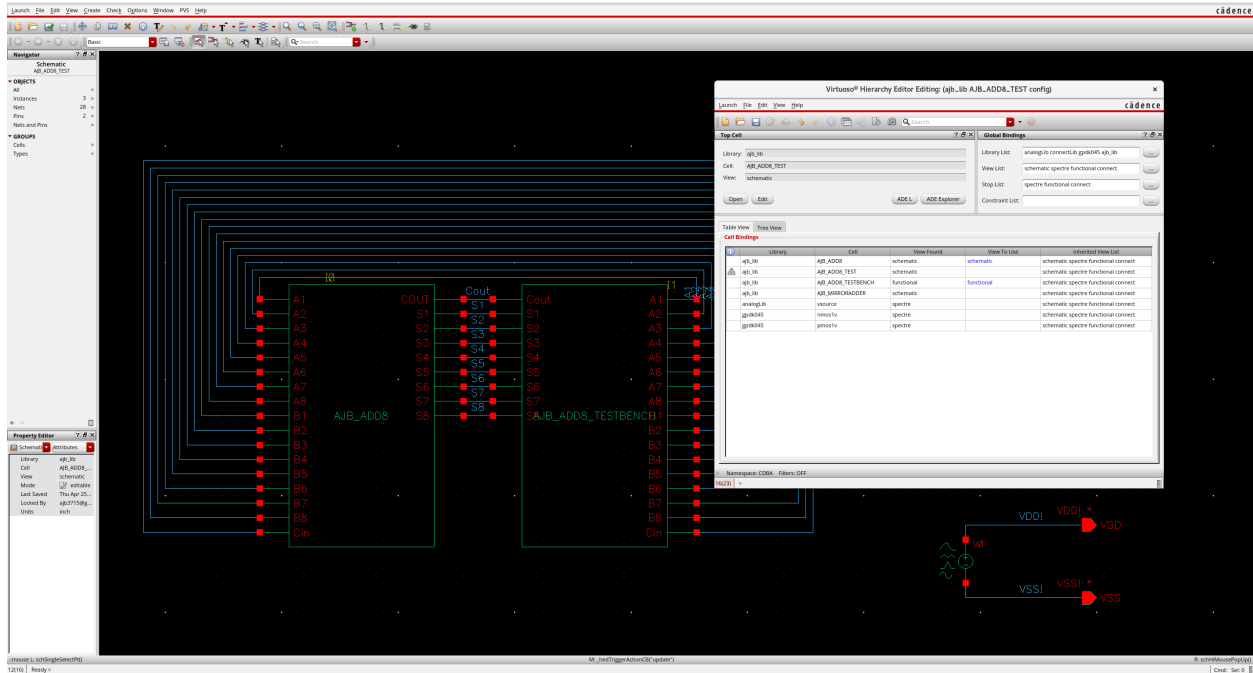


Figure 81: Cell FML_ADD8_TEST AMS

Click on the ADE L button in the hierarchy editor to open the simulation console.

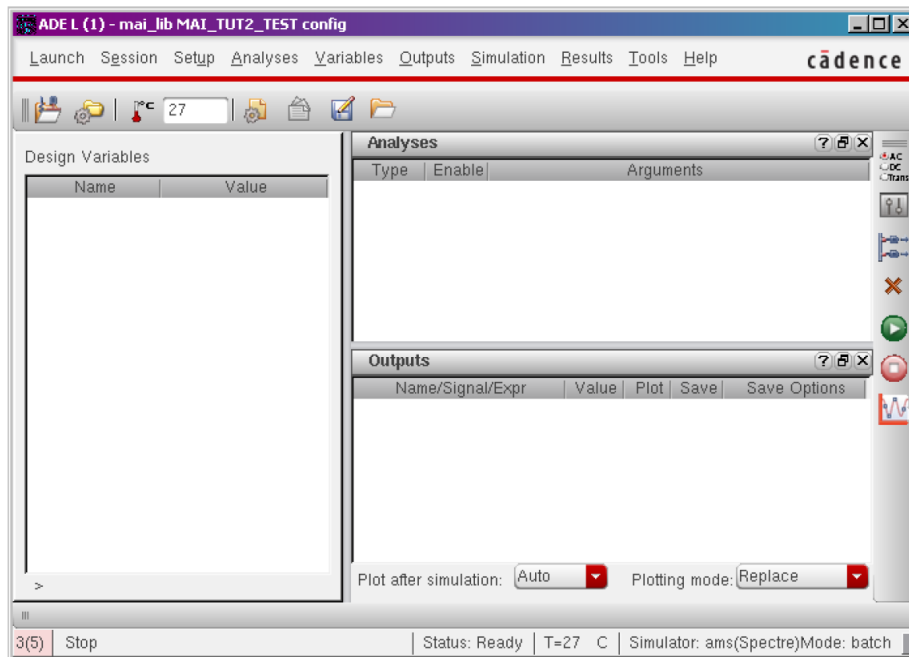


Figure 82: ADE L Simulation Console

We'll now configure various settings to control the simulation. Click **Setup** → **Simulator** to choose the AMS simulator executable to use.

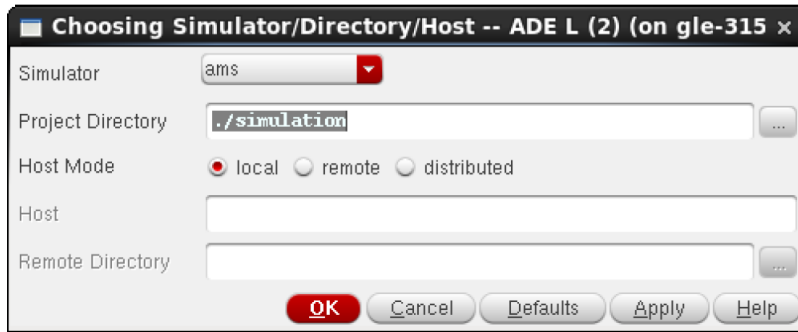


Figure 83: ADE L Simulation Simulator

Click OK. Then, click **Setup** → **Model Libraries** to setup model libraries to use.

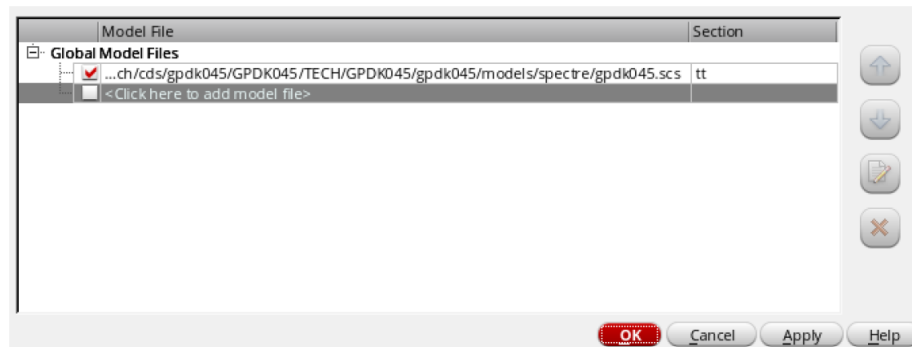


Figure 84: ADE L Setup Model Libraries

Click OK to accept the settings. Click **Setup** → **Connect Rules** to setup the connect rules; connect rules are Verilog -AMS modules used to implement the analog/digital signal conversion between the digital and analog blocks.

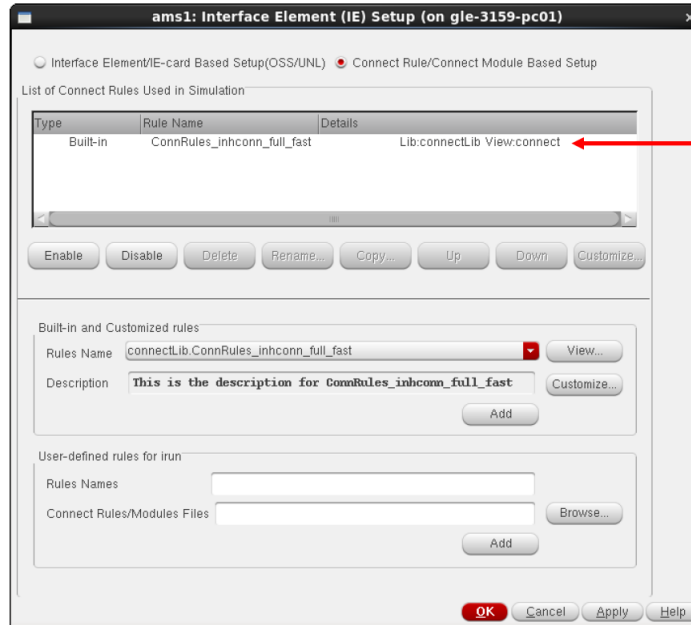


Figure 85: ADE L Setup Connect Rules

As shown in Figure 86, only ConnRules_inhconn_full.fast should be in the list of rules for simulation. Add, delete as necessary, and then click OK to accept the settings.

Click Analyses to setup the analysis as shown.

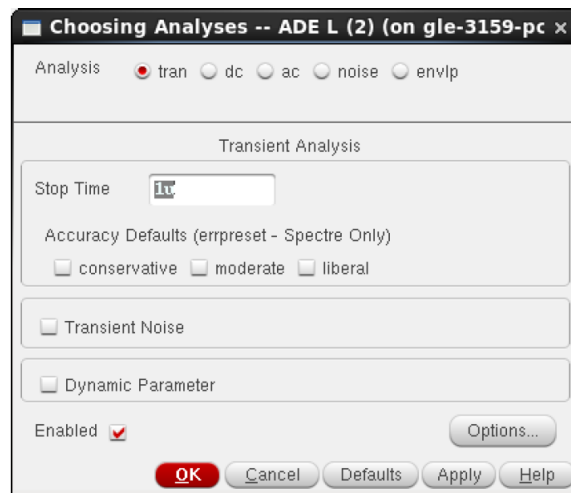


Figure 86: ADE L Analyses

Click OK to accept the settings. Then, click **Outputs** → **To Be Plotted** → **Select On Design** and choose the inputs and outputs on the schematic for plotting.

Click **Simulation** → **Netlist and Run Options** and setup as shown in Figure 87.

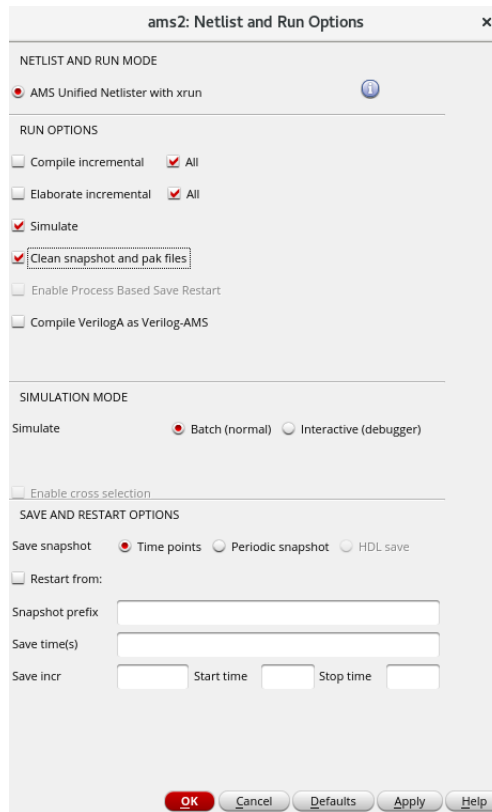


Figure 87: ADE L Analyses

Click **OK** to accept settings. Now click the **Netlist and Run** button (the green arrow button on the right edge of the ADE window) to start the simulation process. The simulation will fail, but a netlist should be generated which is needed for the next step.

Click **Simulation** → **Options** → **AMS Simulator** and choose the Netlister tab.

AMS Options

Main

Netlister

Messages

PLI

SDF

Timing

Low Power

UVM

Miscellaneous

Default Global Signal Declarations

supply0

supply1

ground

gnd!

Netlister Options

Port Connection By

☒ name
☐ order

Port Drilling

☐ all
☐ except top

Netlist using Spectre

☐ Pure Analog Cellviews
☐ Extracted Views

Generate iprobe in

☒ Spectre
☐ netlist.vams
☐ netlist.vams with OOMR

Use symbol for blackboxes bound to external HDLs

☐

in extracted views

☐

Ignore netType property on schematic nets

☐

Run CDF netlister procedure for schematic instance

☐

Enable package importing in AMS Unified Netlister

☐

Check Terminal Number Mismatch

ignore

Check Terminal Direction Mismatch

ignore

Design Vars to be Netlisted as Integer

Include files

s.vams userDisciplines.vams

Maximum number of warnings

Suppress specific netlister info/warnings

Upgrade specific netlister messages to WARNING

Upgrade specific netlister messages to ERROR

Netlist using spectre CDF simInfo

spectre veriloga spice

Netlist instances with sourcefile property as analog

☐

Header text

none

Template header file included

Template header output from script

Global Signals...

OK

Cancel

Defaults

Apply

Help

Figure 88: ADE L AMS Simulator Options

Click **Global Signals** to configure the treatment of global signals.

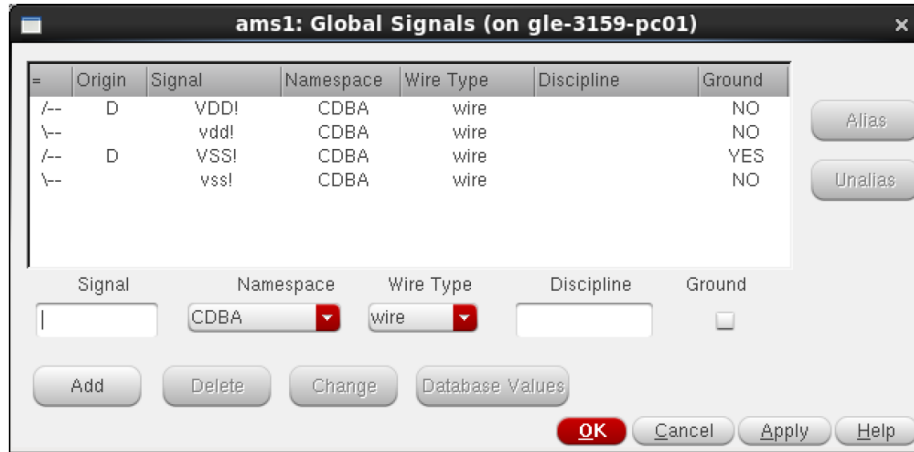


Figure 89: ADE L Simulation Netlist and Run Options

Configure as shown - note the Ground column; you will need to manually add vdd! and vss! (The lower case versions). Note the characters in the “=” column. These indicate signals that are “aliased”. To alias signals, select two signals and then click the **Alias** button. Once the Global Signals are configured, click OK to accept the settings.

Click the **Netlist and Run** button (the green arrow button on the right edge of the ADE window) to start the simulation process.



Figure 90: ADE L Default Waveform Window

Note that while this waveform tool is useful, the SimVision digital centric waveform tool is also available. Click **Tools** → **SimVision Waves** as shown in Figure 91

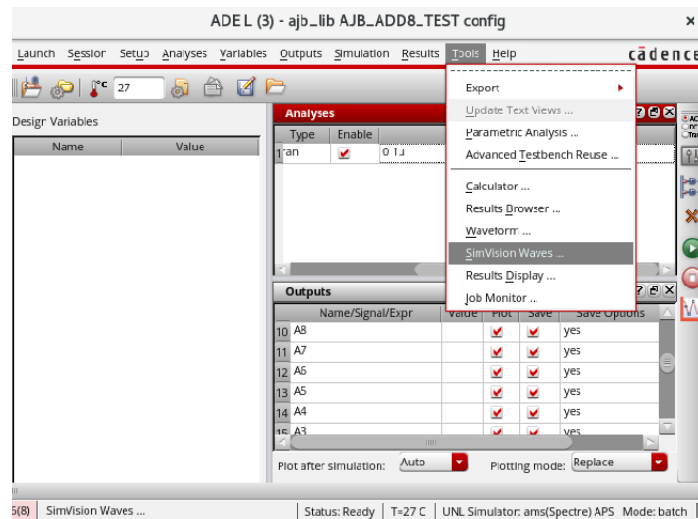


Figure 91: ADE L Open SimVision Waves

SimVision will open as shown. In order for SimVision to know what results data base has to use Select **File** → **Open Database**.

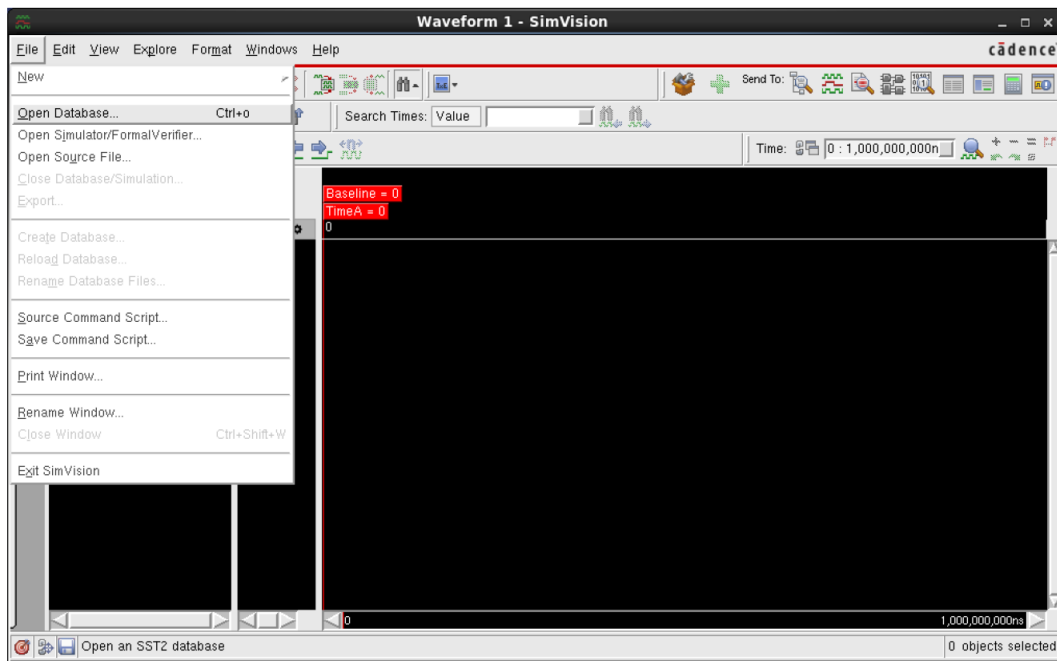


Figure 92: ADE L Open SimVision Waves

Select the database as shown below.

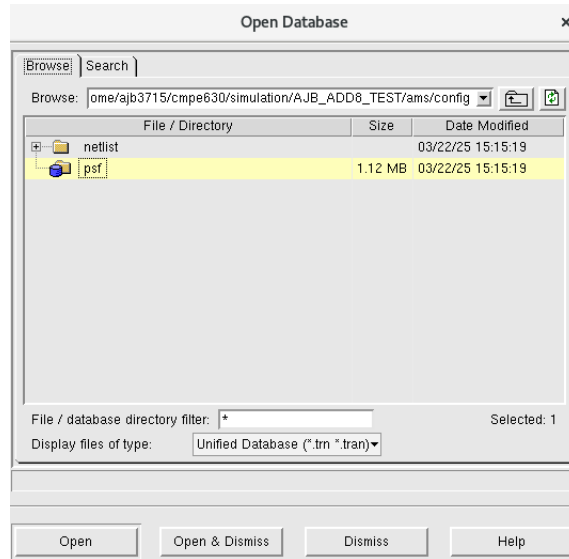


Figure 93: SimVision Select Database

Click **Open & Dismiss** to accept the settings. Now, the database is now opened in SimVision.

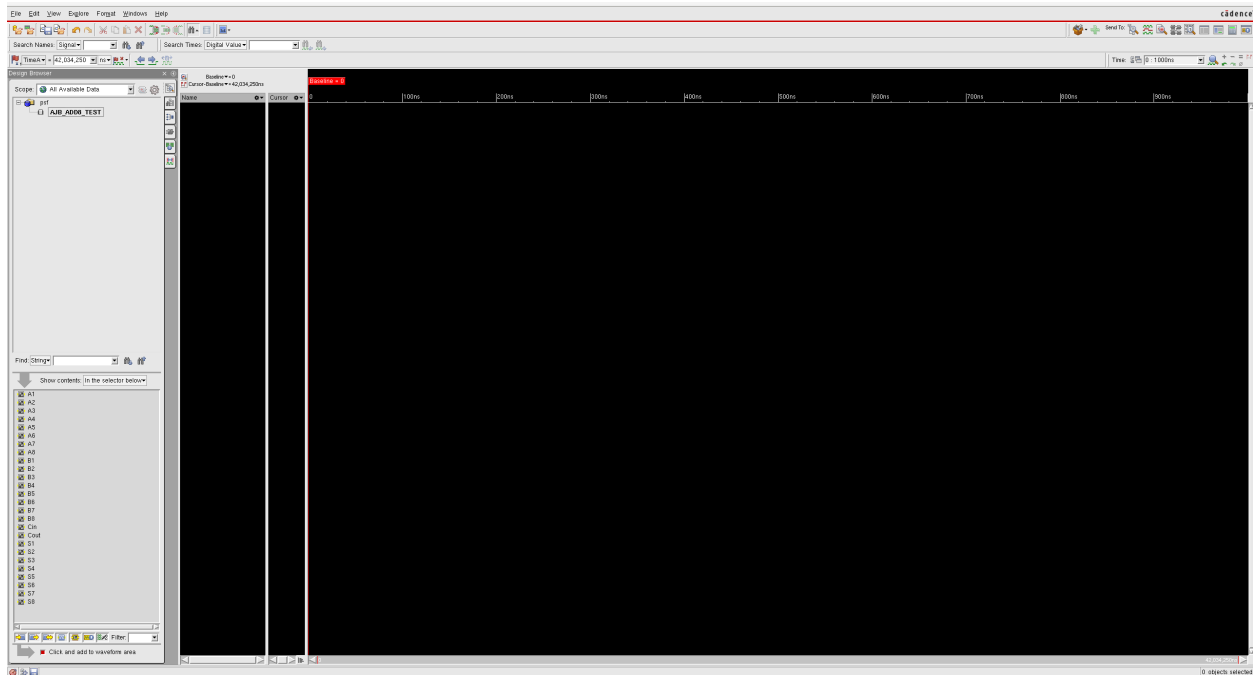


Figure 94: SimVision Database Opened

Modify the SimVision for proper handling of the naalog signals, select **Edit** → **Preferences** and navigate to the **Waveform Window** → **Analog** section and update the options as shown.

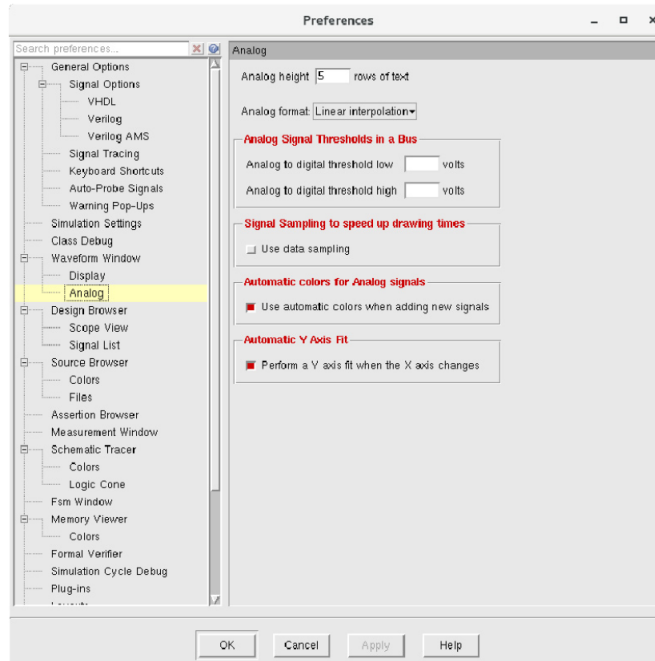


Figure 95: SimVision Preferences

Now add the waveforms to the display window. Right click on the “top” cell, FML_ADD8_TEST, and select the **Send to Waveform Window** option. This will send all available signals in the top cell to the display.

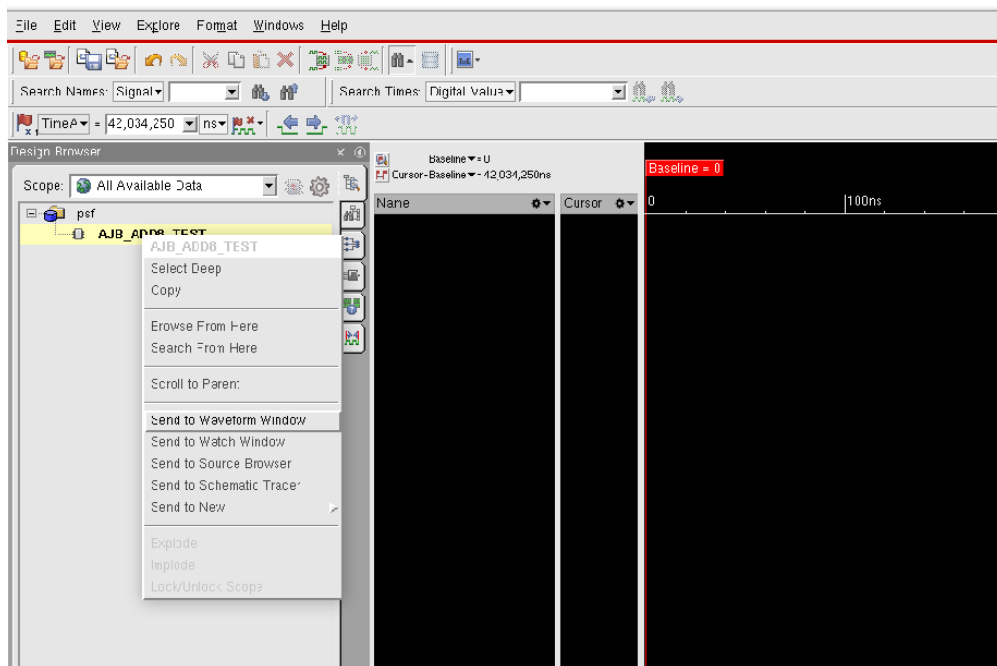


Figure 96: SimVision Add Waveforms

The selected waveforms will display as shown below.

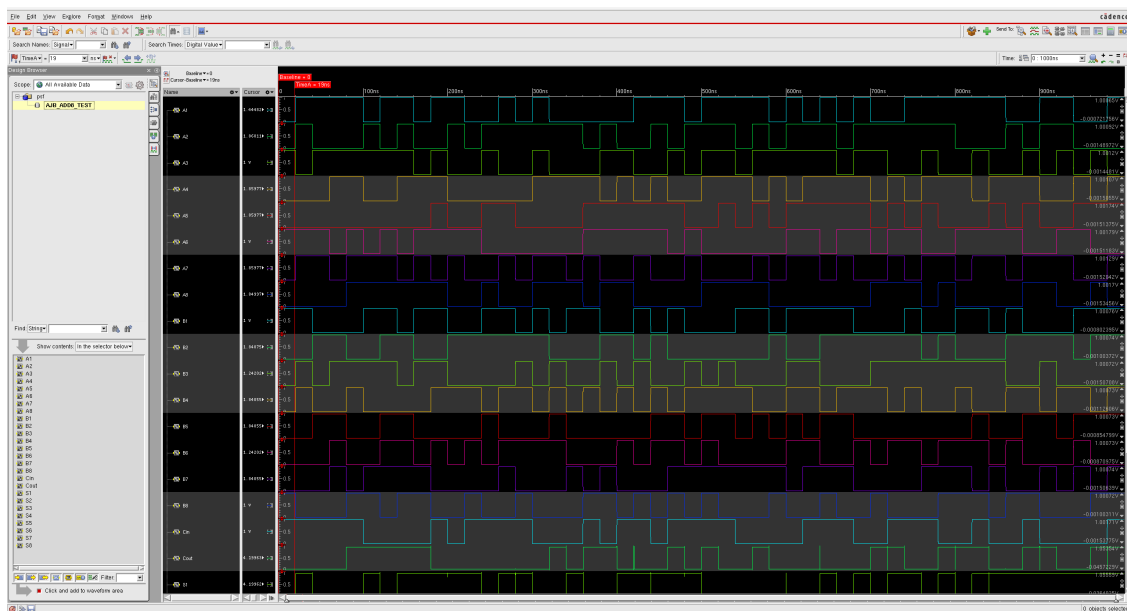


Figure 97: SimVision Waveforms

The display has various options, zoom in and observe the waveform as shown in Figure 98.

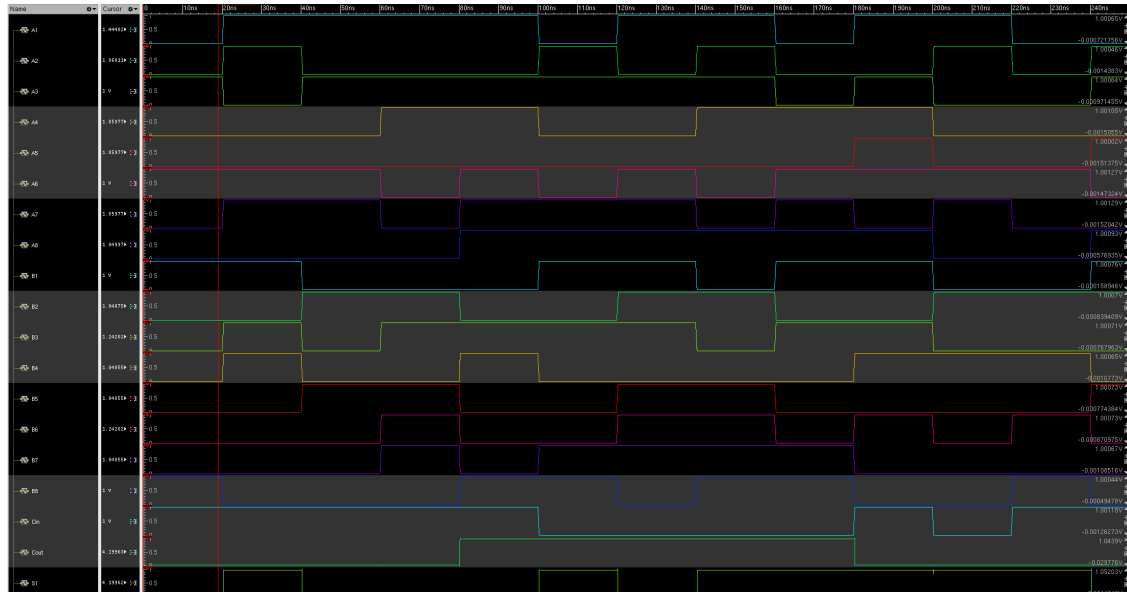


Figure 98: SimVision Waveforms, Zoomed

Congratulations, you've successfully completed your first Mixed-Signal Simulation!

In this tutorial, we used a very simple Verilog HDL testbench. Complex testbenches, which typically include self checking features, could be designed to partially or fully automate circuit testing.

19 Sign-off Sheet

There is no lab report or worksheet for this assignment.

Table 1: Auto Place and Route Tutorial

Students Name:

Section:

Demo	Point Value	Points Earned	Date
P1: DRC , LVS , RC RCX	50		
P2: Mixed-Signal Simulation	50		
Total Points	100		